

Autodesk® Scaleform®

Scaleform CLIK AS3

Scaleform CLIK

Prasad Silva, Matthew Doyle

2.0

2010 8 19

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Autodesk® Scaleform® 4.3

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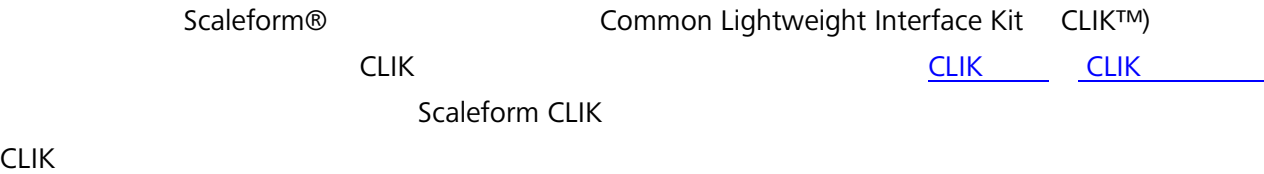
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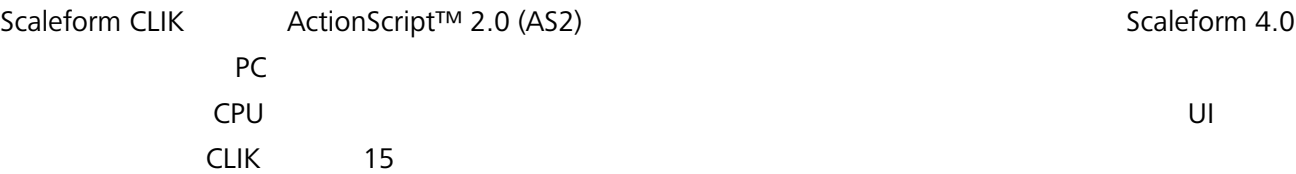
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1



1.1



1.1.1. Scaleform CLIK

UI

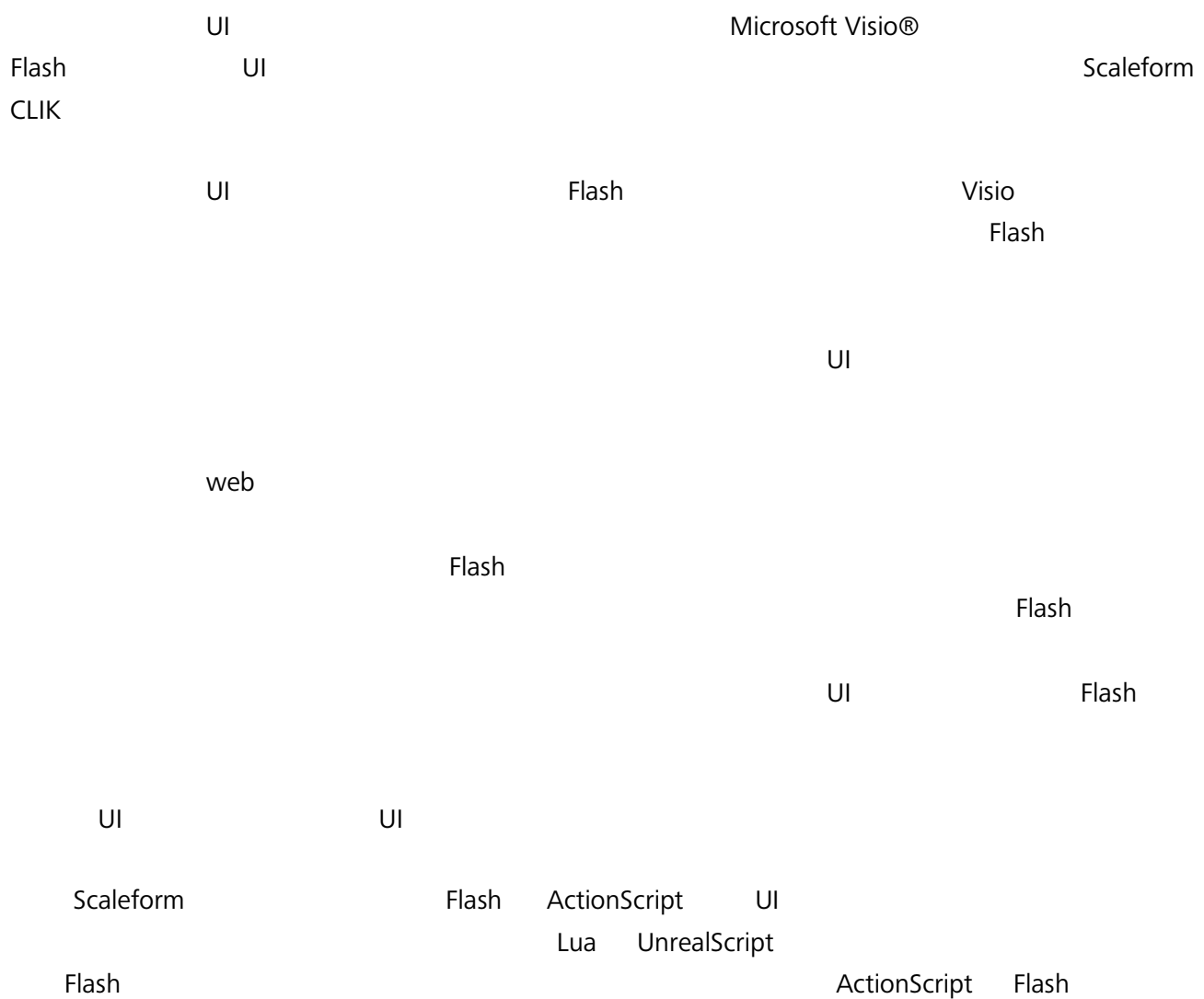
Button	Slider
ButtonBar	StatusIndicator
CheckBox	TileList
RadioButton	Label
TextInput	ScrollingList
TextArea	DropDownMenu
ScrollIndicator	NumericStepper
ScrollBar	

API

InputManager	UIComponent
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IDataProvider
IUIComponent	IList
Constraints	IListItemRenderer

Getting Started with CLIK
Getting Started with CLIK Buttons
CLIK AS3 API Reference
CLIK AS3 User Guide
CLIK Flash Samples
CLIK Video Tutorials

1.2 UI



1.3

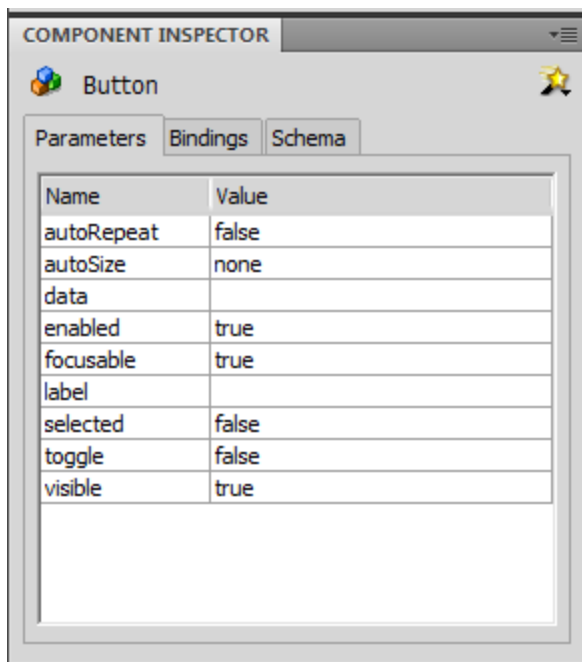
Flash “ ” Flash
Scaleform CLIK
Scaleform gskinner.com

gskinner.com Grant Skinner Grant Skinner Adobe Flash Creative Suite®
4 (CS4) Flash gskinner.com
<http://gskinner.com/blog>

CLIK Flash studio CLIK
Scaleform SDK\Resources\AS3\CLIK\components\CLIK_Components_AS3.fla

1.3.1 屬性檢查

Flash IDE Component Inspector Parameters
CS4 Window Component Inspector
(Shift+F7) Component Inspector “ ” AS



SWF

Flash IDE

CLIK

1.4 框架基礎知識

1.4.1 事件

CLIK

Event() Event ,

- ***type***
- ***target***
- ***currentTarget***
- ***eventPhase*** Event
 EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
 EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

ActionScript 3 Event , Adobe ActionScript 3
http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/Event.html

Scaleform , Scaleform , Extensions.enabled
true() ,Scaleform MouseEventEx, MouseEvent (MouseEvent)
MouseEventEx /
MouseEventEx , ,

1.4.2

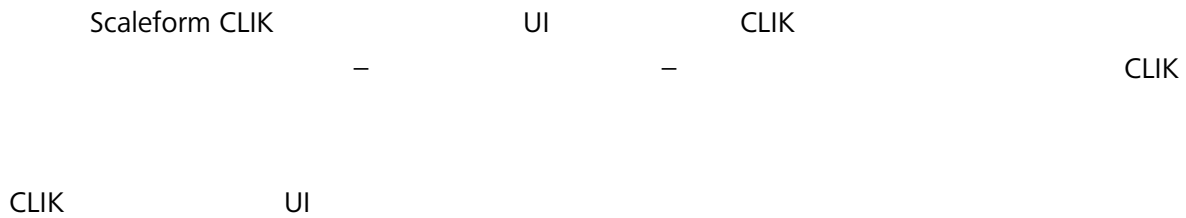
Flash , ,stage
InteractiveObject

Programming

CLIK (Tab) CLIK (Shift+Tab)
CLIK CLIK CLIK CLIK
Flash CLIK CLIK
(Tab) (Shift+Tab)
Enter

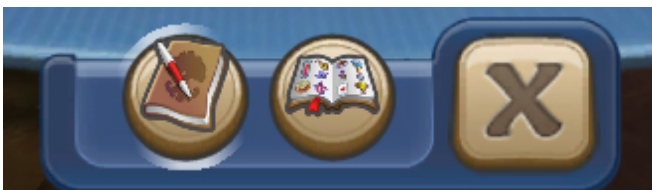
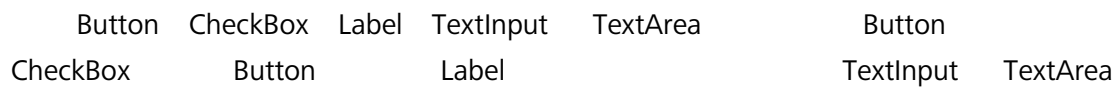
Xbox360 PS3 A X UI CLIK Enter

2

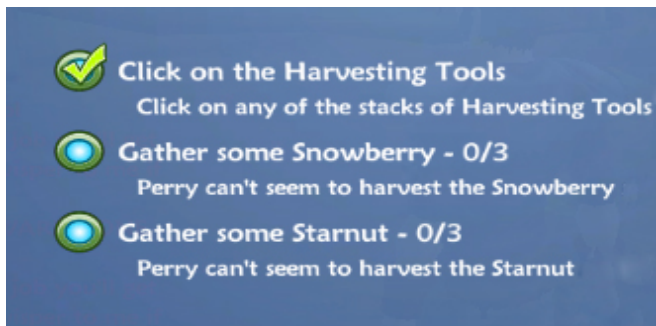


- **User interaction**
- **Component setup** Flash
- **States**
- **Inspectable properties** Flash
- **Events** UI
- **Tips and tricks**

2.1



1 **Free Realms**



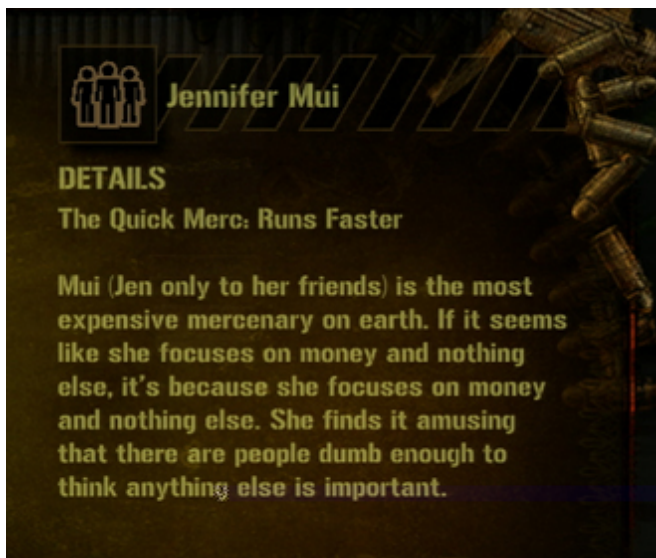
2 *Free Realms* Check box



3 *Free Realms* Label

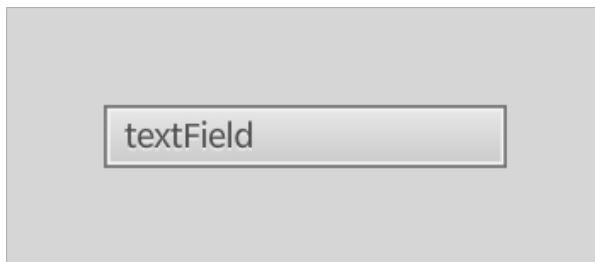


4: *Crysis Warhead* Text input



5 *Mercenaries 2* Text area

2.1.1 Button



6

Button CLIK

Button `scaleform.clik.controls.Button`

textField

ScrollBar

Slider



7 AnimatedButton AnimatedToggleButton ToggleButton

CLIK Button

ToggleButton AnimatedButton AnimatedToggleButton

Button fla

2.1.1.1

2.1.1.2

CLIK Button MovieClip

- **textField:** () TextField
- **focusIndicator:** () MovieClip
 " show " " hide " over

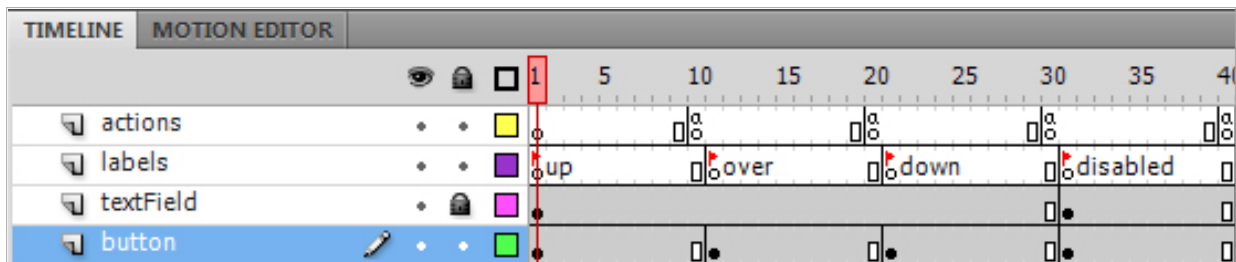
Button

over

2.1.1.3

CLIK

- *up*
- *over*
- *down*
- *disabled*



8 Button

Flash

[CLIK](#)

2.1.1.4

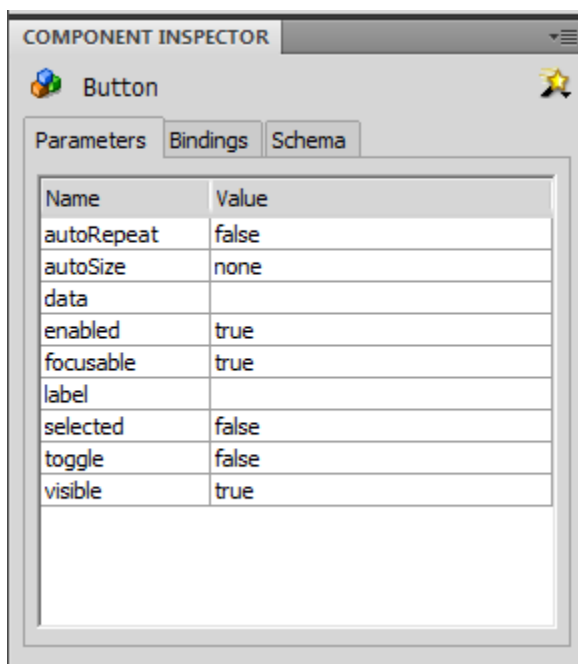


Figure 1: Button component inspectable properties in the CS4 component inspector.

:

autoRepeat	"click"()		
autoSize	autoSize	TextFieldAutoSize.NONE	
Data	ButtonGroup		
enabled	false() ,		
focusable	/	focusable()	false
	tab		
Label			
selected	: toggle() true , , , toggle false, ActionScript		
Toggle	true		
Visible	false		

2.1.1.5

CLIK

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
Event.SELECT	
MouseEvent.ROLL_OVER	mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
MouseEvent.ROLL_OUT	The mouse cursor has rolled out of the button. mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.PRESS	DropDownMenu() controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false – isRepeat: true – autoRepeating ; false –
MouseEvent.DOUBLE_CLICK	doubleClickEnabled true mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.CLICK	controllerIdx: () uint isKeyboard: true - (Keyboard)/

2.1.2.1

CheckBox

2.1.2.2

CLIK CheckBox

MovieClip

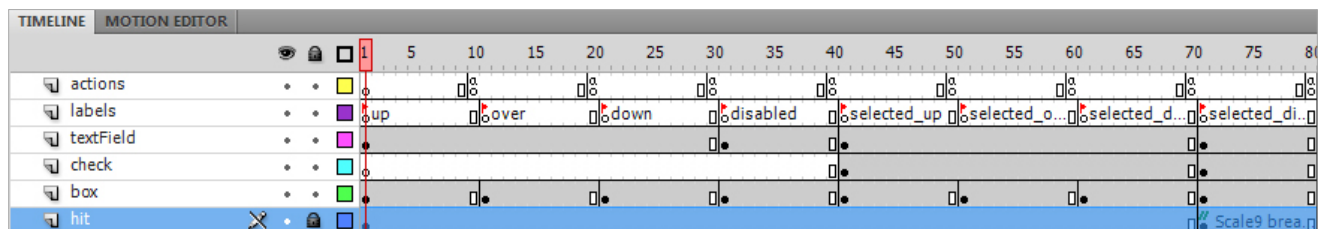
- **textField:** () TextField
- **focusIndicator:** () MovieClip
 MovieClip “ show ” “ hide ”

2.1.2.3

Toggle

CheckBox

- **up**
-
- **down**
- **disabled**
- **selected_up**
-
- **selected_over**
- **selected_down**
- **selected_disabled**



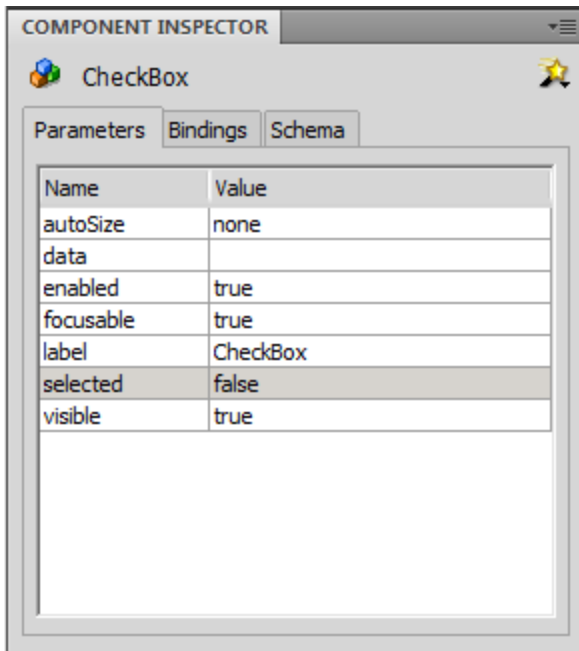
11: CheckBox

CheckBox

Button

[CLIK](#)

2.1.2.4



CheckBox

autoRepeat

Toggle

autoSize			
	autoSize		
	autoSize=TextFieldAutoSize.NONE		
Data	ButtonGroup		
enabled	false,		
focusable	/	focusable()	false
	tab		
Label			
selected	:		
	toggle()	true ,	
	, , toggle	false,	
	ActionScript		
Visible	false		

2.1.2.5

Event() Event ,

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

CheckBox

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
Event.SELECT	
ComponentEvent.STATE_CHANGE	
MouseEvent.ROLL_OVER	mouseIdx: (uint, Scaleform, MouseEventEx) buttonIdx: (MouseEventEx)
MouseEvent.ROLL_OUT	mouseIdx: (uint, Scaleform, MouseEventEx) buttonIdx: (MouseEventEx)
ButtonEvent.PRESS	controllerIdx: (uint) isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -

MouseEvent.DOUBLE_CLICK	<i>doubleClickEnabled</i>	true
	mouseIdx:	(
	) uint	Scaleform,
	MouseEventEx	buttonIdx:
	(	Scaleform,
	MouseEventEx	
ButtonEvent.CLICK	controllerIdx:	(
	) uint	
	isKeyboard: true -	(Keyboard)/
	(Gamepad) ; false -	
	isRepeat: true -	
	autoRepeating ; false -	
ButtonEvent.DRAG_OVER	controllerIdx:	(
	) uint	
ButtonEvent.DRAG_OUT	controllerIdx:	(
	) uint	
ButtonEvent.RELEASE_OUTSIDE	controllerIdx:	(
	) uint	

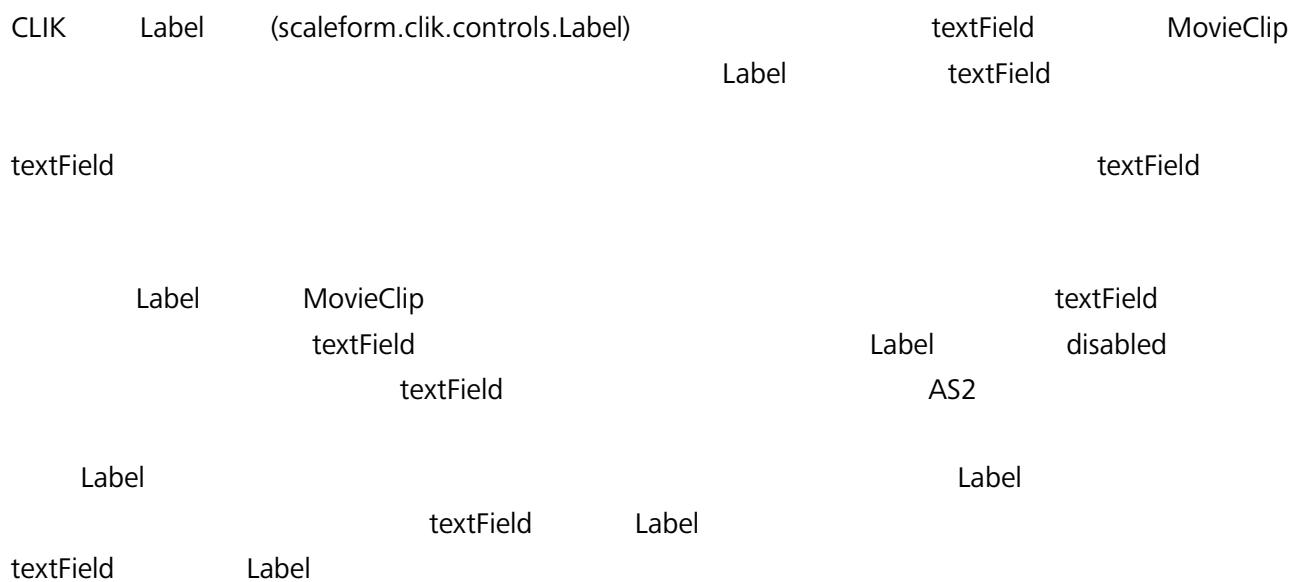
CheckBox Toggle

```
myCheckBox.addEventListener(Event.SELECT, onCheckBoxToggle);
function onCheckBoxToggle(e:Event):void {
    // Do something
}
```

2.1.3 Label



12:



2.1.3.1

Label

2.1.3.2

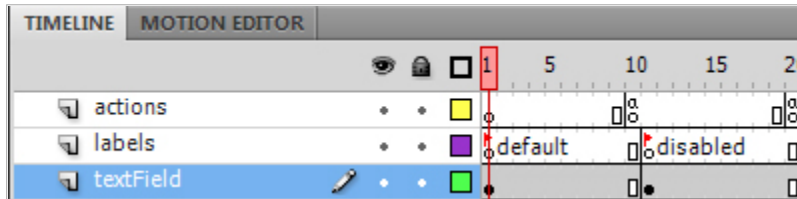
CLIK Label MovieClip

- **textField:** TextField Label

2.1.3.3

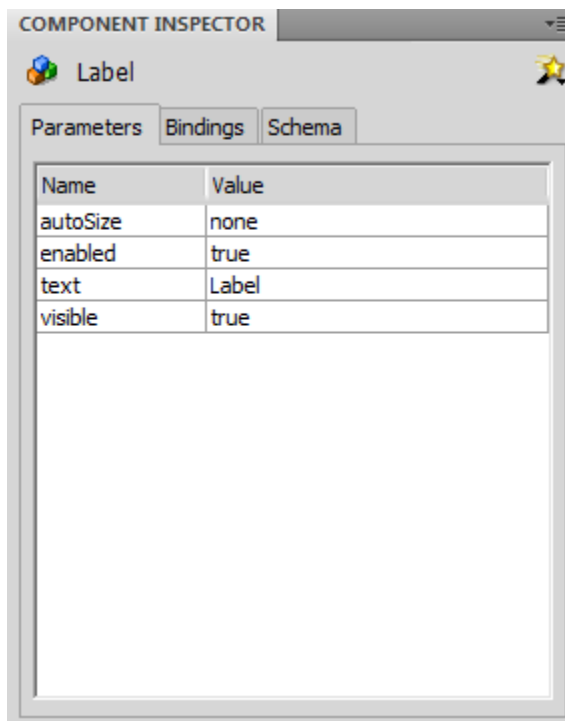
CLIK Label

- **default** enabled
- **disabled**



13: Label

2.1.3.4



Label

Text			
Visible	false,		
Enabled	false,		
autoSize	(Label)	autoSize	autoSize=TextFieldAutoSize.NONE

2.1.3.5

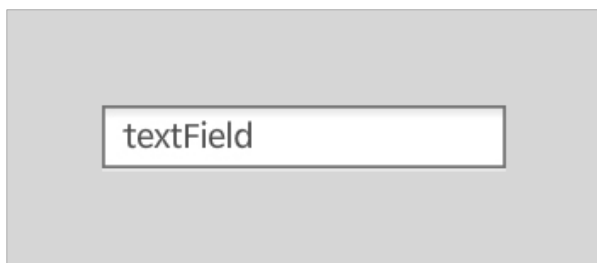
Event() Event ,

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

Label

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		

2.1.4 TextInput



14: TextInput.

TextInput (scaleform.clk.controls.TextInput)	textField
--	-----------

Label	textField	textField
-------	-----------	-----------

HTML TextInput

textField

textField	Label	Label
-----------	-------	-------

Label Label

Label

textFields	TextInput	UI
------------	-----------	----

TextInput UI

PC	TextInput	textField
----	-----------	-----------

TextInput	textField
-----------	-----------

textField

UI

```

        TextInput disabled textField
    TextInput
    TextInput AS2
    TextInput
    Flash Studio
    TextInput

    TextInput textField,
    (Shift+Arrows) (Shift+Delete) HTML
    (Shift+Insert) (Ctrl+Insert)

    " " actAsButton
    "over" "out"
    " " " " " "
    actAsButton "over"/"out"
    " "
    (0xAAAAAA) " " "
    "

```

2.1.4.1

```

    TextInput textField
    textField

```

2.1.4.2

```

    CLIK TextInput MovieClip

```

- **textField**: textField

2.1.4.3 States

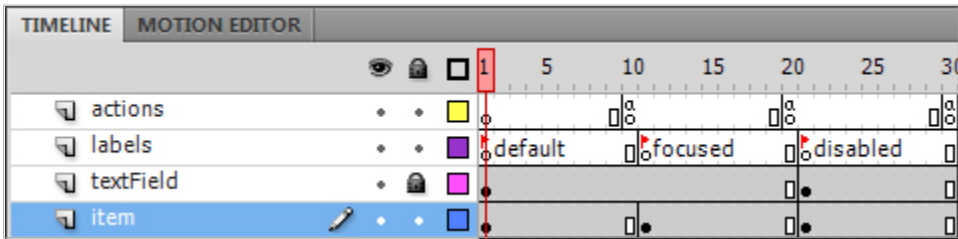
```

    CLIK TextInput disabled

```

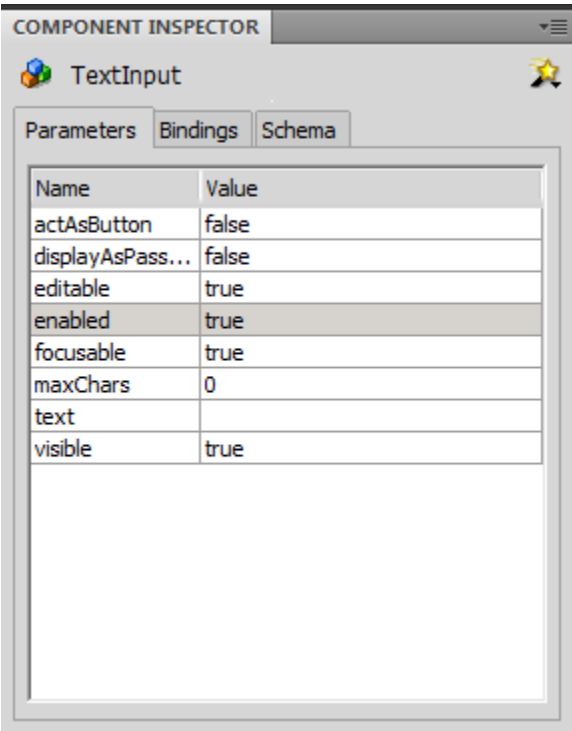
- **default** enabled
- **focused** textField

- *disabled*



15: TextInput

2.1.4.4



TextInput :

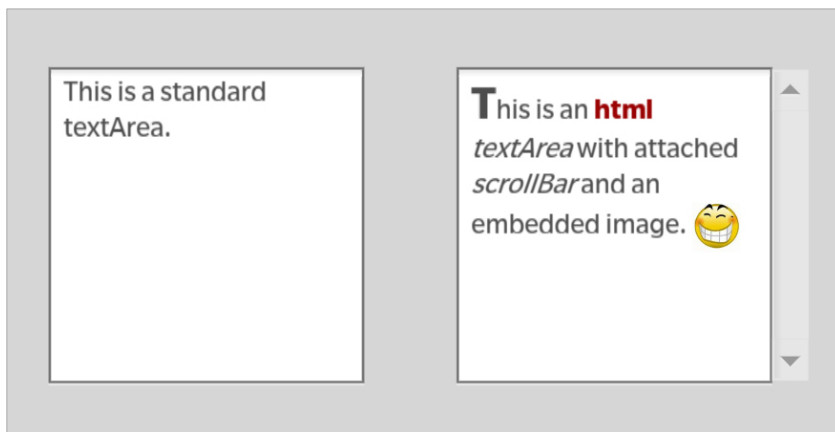
actAsButton	" " " "	tab
displayAsPassword	true, textField '*' , textField ,	
Editable	false, TextInput	
Enabled	Disables the component if set to false.	
Focusable	/ false tab	focusable()


```
buttonIdx:
    )      Scaleform,
MouseEventEx
```

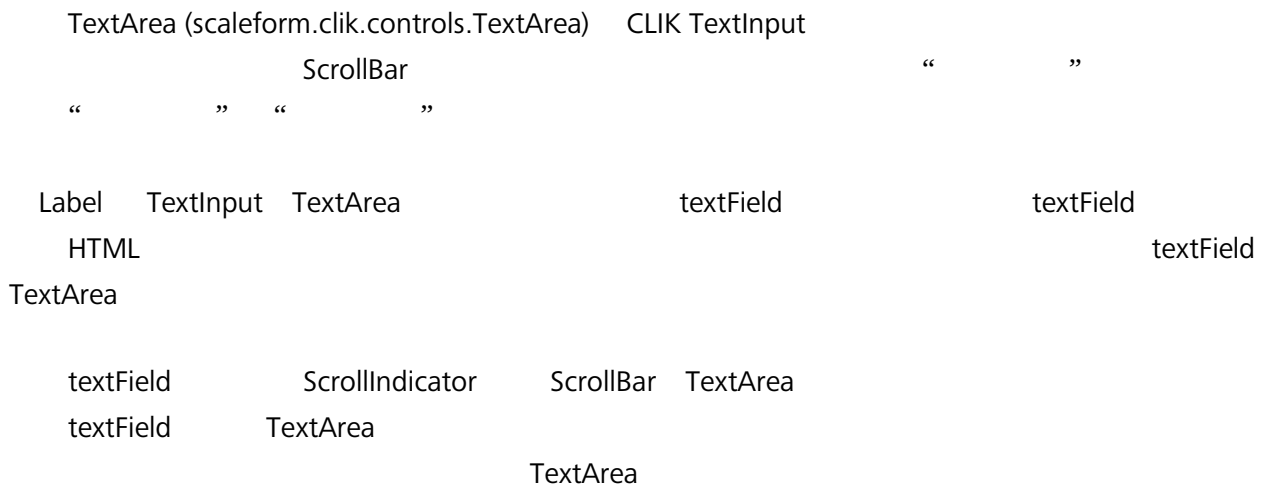
textField

```
myTextInput.addEventListener(Event.CHANGE, onTextChange);
function onTextChange(e:Event):void {
    trace("Latest text: " + e.target.text);
}
```

2.1.5 TextArea



16: **TextArea.**



2.1.5.1

TextArea

textField

textField

2.1.5.2

CLIK TextArea

MovieClip

- **textField**: TextField

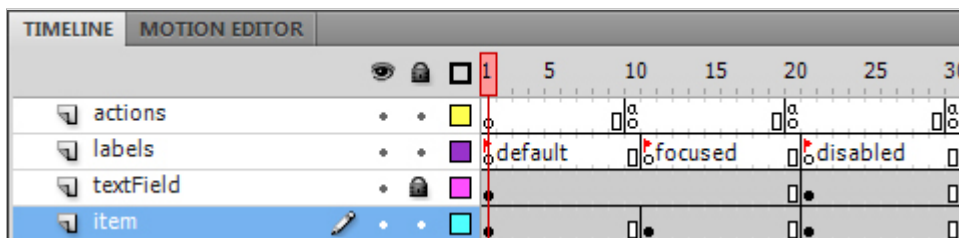
2.1.5.3

TextInput

TextArea

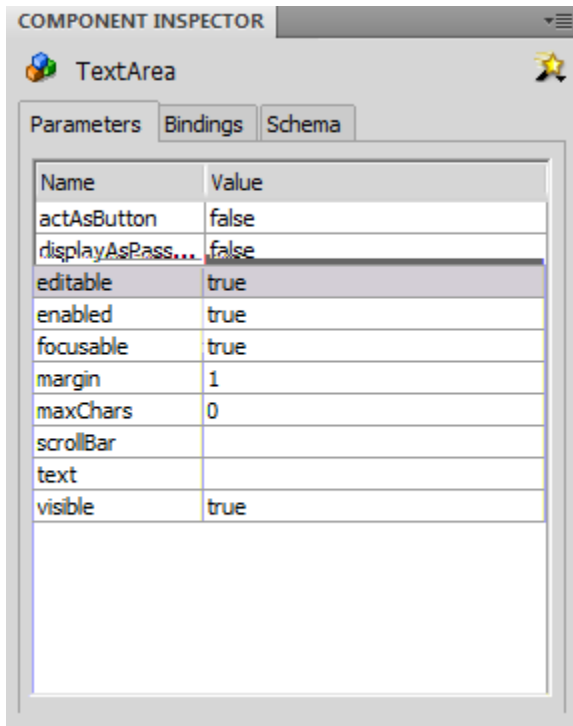
disabled

- **default** enabled
- **focused** textField
- **disabled**



17: TextArea

2.1.5.4



TextArea

TextInput

CLIK ScrollBar

2.4

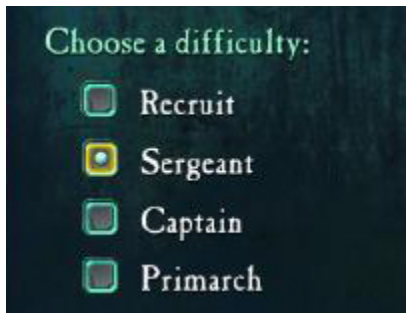
actAsButton	true,	TextInput	,	,
	rollOver	rollOut	tab	,TextInput
		,		
displayAsPassword	true,	textField	'*'	textField
			,	
Editable		false	TextInput	
Enabled		false,		
focusable	/		focusable()	false
		tab		
maxChars			textField	
scrollBar	CLIK ScrollBar		ScrollBar	ID
		TextArea		
scrollPolicy	" auto"			
	" on"	ScrollBar		" off"
		ScrollBar	ScrollBar	
Text	textField			

TextArea

```
myTextArea.addEventListener(Event.SCROLL, onTextScroll);  
function onTextScroll(e:Event):void {  
    // Do something  
}
```

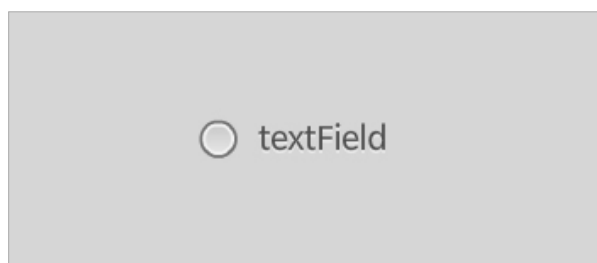
2.2

RadioButton ButtonGroup ButtonBar ButtonGroup
Button ButtonBar
RadioButton ButtonGroup



18: *Dawn of War II*

2.2.1 RadioButton



19: **RadioButton**

RadioButton (scaleform.clik.controls.RadioButton)

CLIK CheckBox ButtonGroup

CheckBox
ButtonGroup

ButtonGroup

2.2.1.1

2.2.1.2

CLIK RadioButton MovieClip

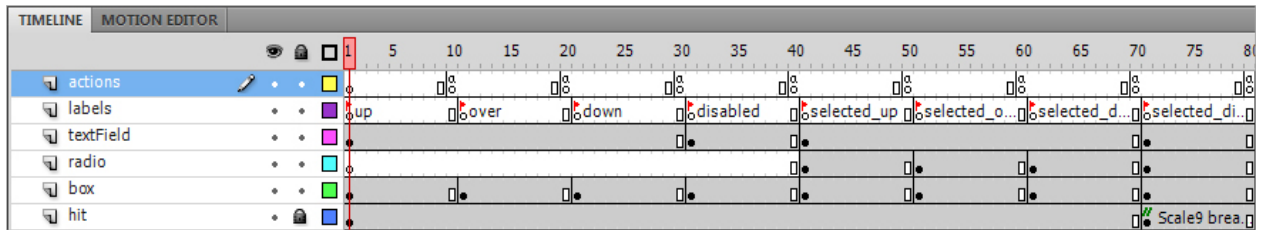
MovieClip

- **textField** () TextField
- **focusIndicator** MovieClip MovieClip
MovieClip “ show ” “ hide ”

2.2.1.3

CheckBox

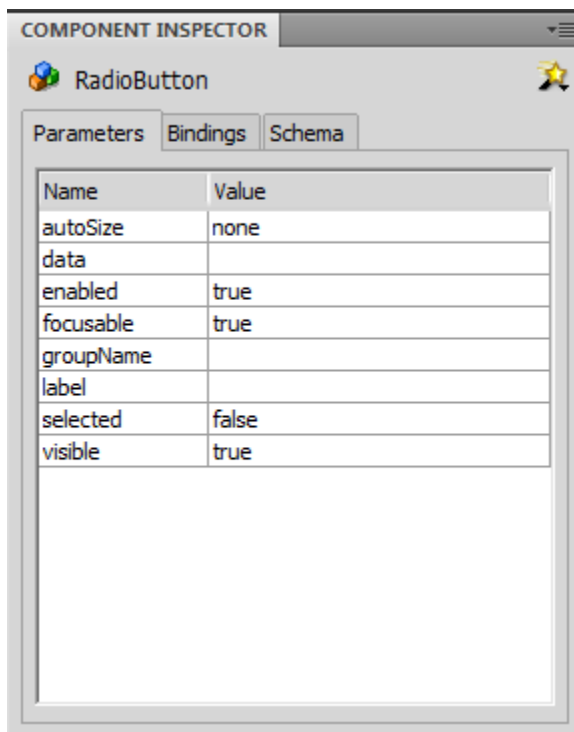
- *up*
-
- *over*
- *down*
- *disabled*
- *selected_up*
-
- *selected_over*
- *selected_down*
- *selected_disabled*



20: RadioButton

RadioButton Button
 RadioButton [CLIK](#)

2.2.1.4



Button RadioButton
 disableFocus Toggle

autoSize	
	autoSize
	autoSize=TextFieldAutoSize.NONE
Data	ButtonGroup
enabled	false,

focusable	/	focusable()	false
	tab		
groupName	ButtonGroup	, ,	RadioButton
	RadioButton	, ButtonGroup	
	RadioButton	, RadioButton	_root
	_root	ButtonGroup	
	RadioButton		
Label			
selected		:	
	toggle()	true ,	
	, , toggle	false,	ActionScript

2.2.1.5

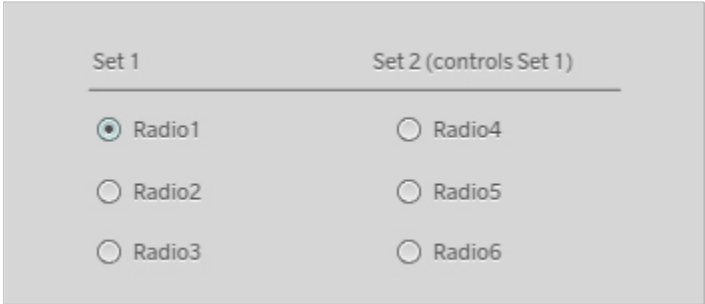
	Event()	Event	,
• type • target • currentTarget • eventPhase • bubbles • cancelable	Event	EventPhase.CAPTURING_PHASE	EventPhase.AT_TARGET
	EventPhase.BUBBLING_PHASE		
	RadioButton		
ComponentEvent.SHOW	visible()	true	
ComponentEvent.HIDE	visible	false	
FocusHandlerEvent.FOCUS_IN			
FocusHandlerEvent.FOCUS_OUT			
Event.SELECT			
ComponentEvent.STATE_CHANGE			
MouseEvent.ROLL_OVER	mouseIdx:	(	

	) uint MouseEventEx buttonIdx: () Scaleform, MouseEventEx
MouseEvent.ROLL_OUT	mouseIdx: ()) uint MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.PRESS	controllerIdx: ()) uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -
MouseEvent.DOUBLE_CLICK	<i>doubleClickEnabled</i> true mouseIdx: ()) uint MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.CLICK	controllerIdx: ()) uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false - isRepeat: true - autoRepeating ; false -
ButtonEvent.DRAG_OVER	controllerIdx: ()) uint
ButtonEvent.DRAG_OUT	

	controllerIdx:	(
	) uint	
ButtonEvent.RELEASE_OUTSIDE		
	controllerIdx:	(
	) uint	
RadioButton		

```
myRadio.addEventListener(Event.SELECT, onRadioToggle);
function onRadioToggle(e:Event):void {
    // Do something
}
```

2.2.2 ButtonGroup



21: ButtonGroup.

CLIK ButtonGroup (scaleform.clik.controls.ButtonGroup)



2.2.2.1

RadioButton

2.2.2.2

CLIK ButtonGroup MovieClip

2.2.2.3

ButtonGroup

2.2.2.4

ButtonGroup

2.2.2.5

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

ButtonGroup

Event.CHANGE

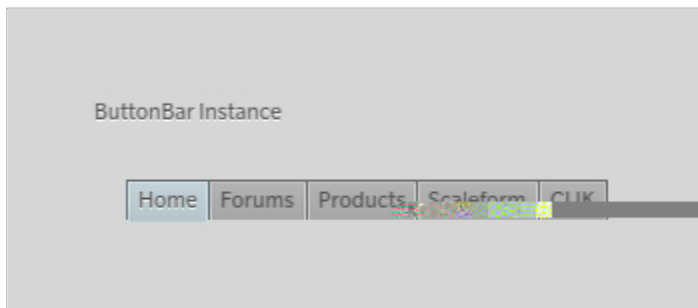
ButtonEvent.CLICK

target: CLIK Button

ButtonGroup

```
myGroup.addEventListener(Event.CHANGE, onNewSelection);
function onNewSelection(e:Event):void {
    if (e.item == myRadio) {
        // Do something
    }
}
```


2.2.3 ButtonBar



22: ButtonBar.

ButtonBar (scaleform.clik.controls.ButtonBar) ButtonGroup
dataProvider dataProvider
tab UI

```
buttonBar.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

2.2.3.1

ButtonGroup

2.2.3.2

CLIK ButtonBar MovieClip

2.2.3.3

CLIK

2.2.3.4

- **target**
- **currentTarget**
- **eventPhase**
- **bubbles**
- **cancelable**

Event

EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET

EventPhase.BUBBLING_PHASE

ButtonBar

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
IndexEvent.INDEX_CHANGE	index:ButtonBar int -1() 1 lastIndex:ButtonBar int -1(1)	AS3 Object
ButtonBarEvent.BUTTON_SELECT	index:ButtonBar Int -1() 1 renderer:ButotnBar Button Button	

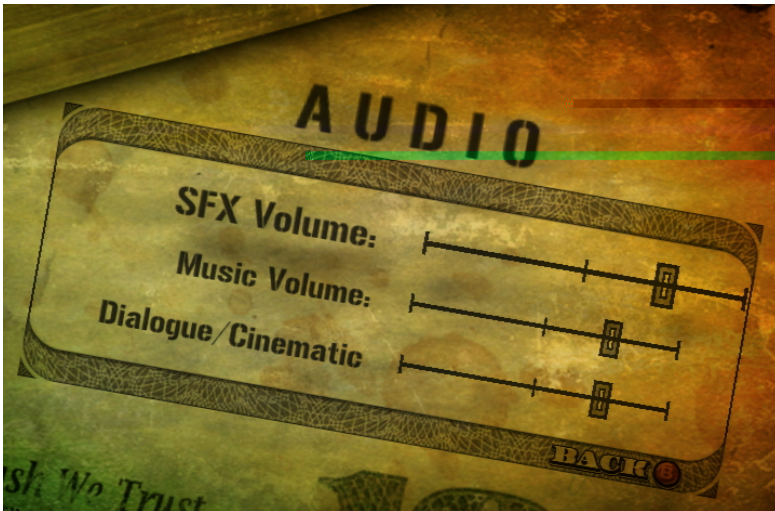
```

myBar.addEventListener(ButtonBarEvent.BUTTON_SELECT, onItemClick);
function onItemClick(e:ButtonBarEvent) {
    processData(e.renderer.data);
    // Do something
}

```

2.3

ScrollIndicator ScrollBar Slider ScrollIndicator
ScrollBar
Slider



23: *Mercenaries 2*

2.3.1 ScrollIndicator



24: **ScrollIndicator.**

CLIK ScrollIndicator (scaleform.clik.controls.ScrollIndicator)
textField
ID
TextArea

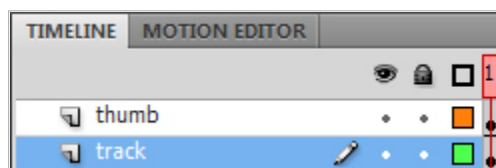
2.3.1.1

2.3.1.2

CLIK ScrollIndicator MovieClip

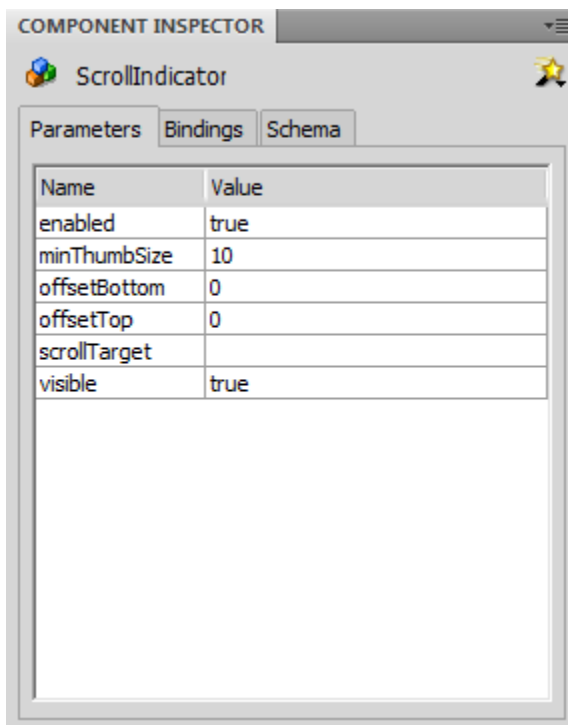
- ***thumb***: CLIK Button
- ***track*** MovieClip

2.3.1.3



25: **ScrollIndicator**

2.3.1.4



Enabled	false,		
offsetTop			
offsetBottom			
scrollTarget	TextArea textField	textField	ScrollIndicator
Visible	false		

2.3.1.5

Event() Event ,

- *type*
-
- *target*
- *currentTarget* Event
- *eventPhase* EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- *bubbles*
- *cancelable*

ScrollIndicator

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
Event.SCROLL	ScrollIndicator	

```
mySI.addEventListener(Event.SCROLL, onTextScroll);
function onTextScroll(e:Event) {
    trace("mySI.position: " + e.target.position);
    // Do something
}
```

2.3.2 ScrollBar



26: ScrollBar.

CLIK ScrollBar(scaleform.clik.controls.ScrollBar)

2.3.2.1

ScrollBar

trackMode
(Shift)

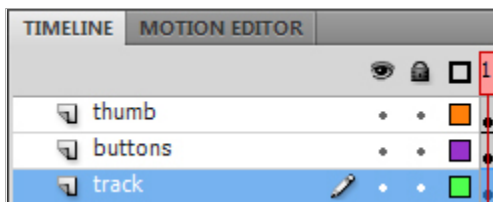
trackMode

2.3.2.2

CLIK ScrollBar MovieClip

- ***upArrow*** CLIK
- ***downArrow***: CLIK
- ***thumb***: CLIK
- ***track***: CLIK

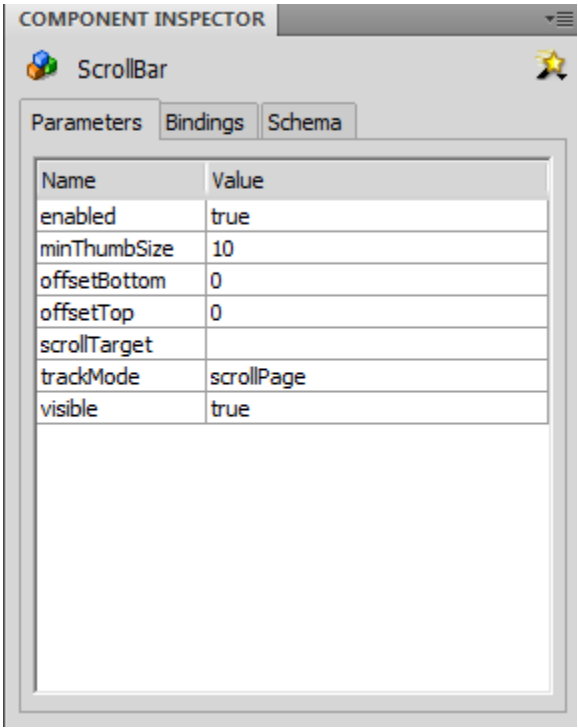
2.3.2.3



27: ScrollBar

ScrollBar ScrollIndicator

2.3.2.4



, trackMode:

Enabled	false,
offsetTop	
offsetBottom	
scrollTarget	TextArea textField
trackMode	scrollPage scrollToCursor
Visible	false

2.3.2.5

Event() Event ,

- *type*
- *target*
- *currentTarget* Event

- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

ScrollBar

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
Event.SCROLL	ScrollBar	

```
mySB.addEventListener(Event.SCROLL, onTextScroll);
function onTextScroll(e:Event) {
    trace("mySB.position: " + e.target.position);
    // Do something
}
```

2.3.3 Slider



28: Slider.

Slider(scaleform.clik.controls.Slider)

2.3.3.1

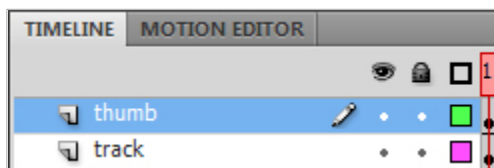
home end

2.3.3.2

CLIK Slider MovieClip

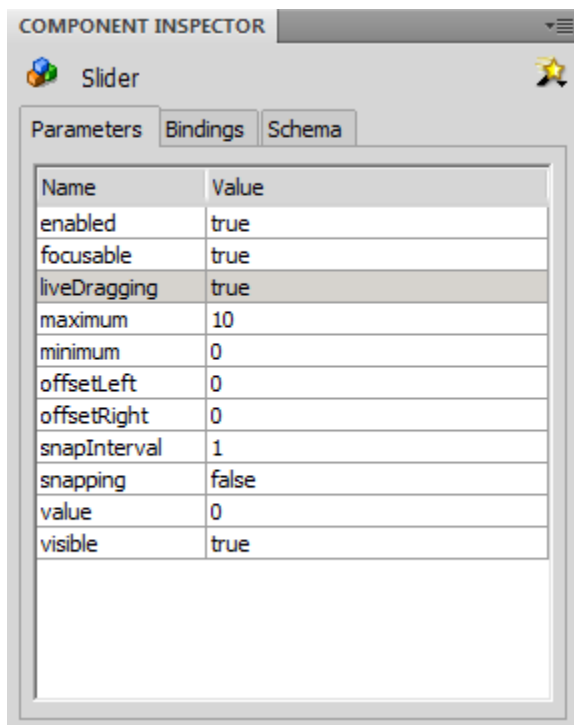
- **thumb** CLIK
- **Track** CLIK

2.3.3.3



29: Slider

2.3.3.4



Slider

Enabled

false,

Focusable	/	focusable()	false
Value	Slider		
Minimum			
Maximum			
Snapping	true		
snapInterval		false	
liveDragging	true	Slider	
offsetLeft			
offsetRight			
Visible	false		

2.3.3.5

Event() Event ,

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

Slider

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
SliderEvent.VALUE_CHANGE	Slider's	

Slider

```
mySlider.addListener(SliderEvent.VALUE_CHANGE, onValueChange);
function onValueChange(e:SliderEvent) {
    trace("mySlider.value: " + e.target.value);
    // Do something
}
```

```
}
```

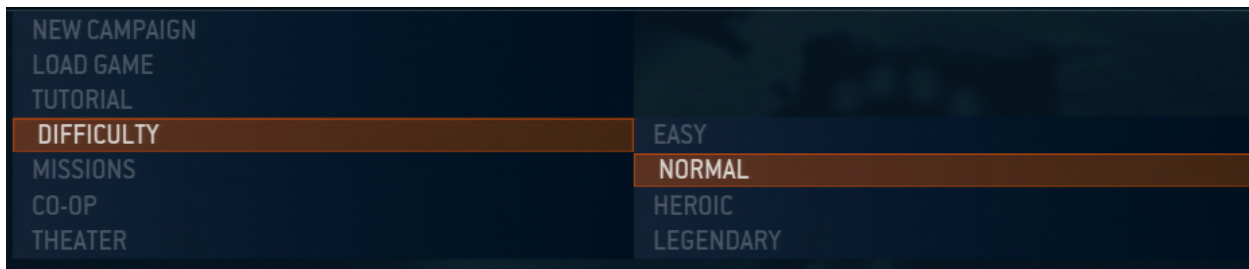
2.4

CLIK NumericStepper OptionStepper ScrollingList TileList DropdownMenu
 NumericStepper DataProvider ListItemRenderer
 ScrollingList TileList

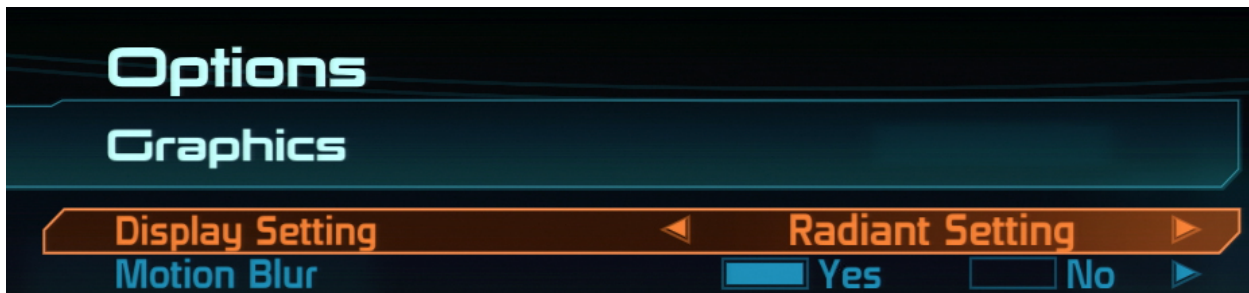
NumericStepper OptionStepper ScrollingList TileList
 ScrollIndicator ScrollBar DropdownMenu
 ScrollingList TileList



30: *Mercenaries 2*



31: *Halo Wars* 'Difficulty Settings'



32: *Mass Effect* 'Display Setting'

2.4.1 NumericStepper



33: **Numeric Stepper.**

NumericStepper (scaleform.clik.controls.NumericStepper)

2.4.1.1

NumericStepper

(Home)

(End)

2.4.1.2

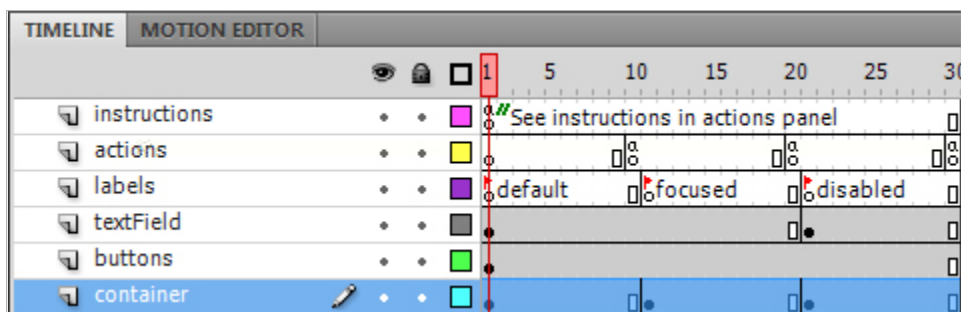
CLIK NumericStepper MovieClip

- **textField**: TextField
- **nextBtn**: CLIK Button
- **prevBtn**: CLIK Button

2.4.1.3

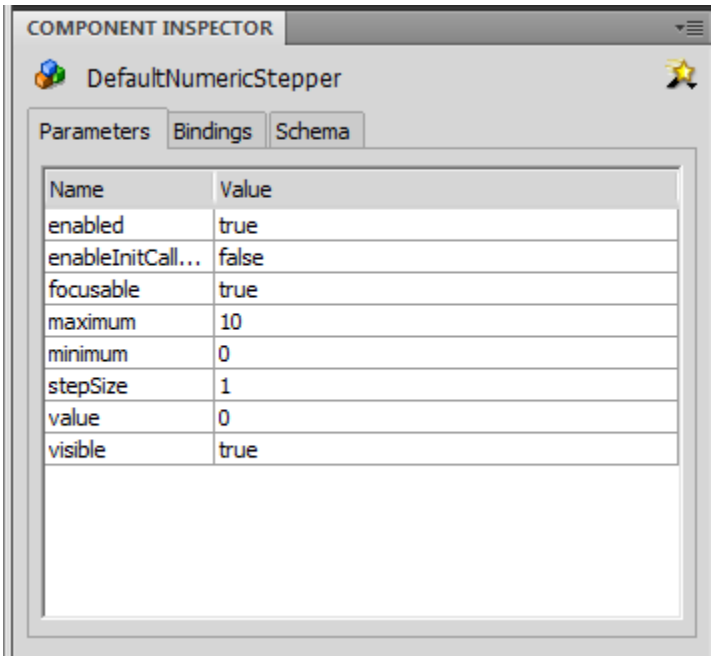
NumericStepper

- **default** enabled
- **focused** textField
- **disabled**



34: NumericStepper

2.4.1.4



NumericStepper

MovieClip

Enabled	false,		
Focusable	/	focusable()	false
tab			
minimum	NumericStepper		
maximum	NumericStepper		
stepSize	NumericStepper	,	/ NumericStepper
Value	Slider		
Visible	false		

2.4.1.5

Event() Event ,

- *type*
- *target*
- *currentTarget* Event
- *eventPhase* EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE

- *bubbles*
- *cancelable*

NumericStepper

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
IndexEvent.INDEX_CHANGE	NumericStepper	
	index:NumericStepper	int -
	1()	1
	lastIndex:NumericStepper	int
	-1()	
	1	
	data: dataProvider	AS3 Object

NumericStepper

```
myNS.addEventListener(IndexEvent.INDEX_CHANGE, onValueChange);
function onValueChange(e:IndexEvent) {
    trace("myNS.value: " + e.target.value);
    // Do something
}
```

2.4.2 OptionStepper



35: **OptionStepper.**

OptionStepper (scaleform.clik.controls.OptionStepper), NumericStepper
 dataProvider

OptionStepper

dataProvider

```
optionStepper.dataProvider = new DataProvider(["item1", "item2", "item3",  
"item4"]);
```

2.4.2.1

NumericStepper

OptionStepper

(Home) (End)

dataProvider

2.4.2.2

CLIK NumericStepper

MovieClip

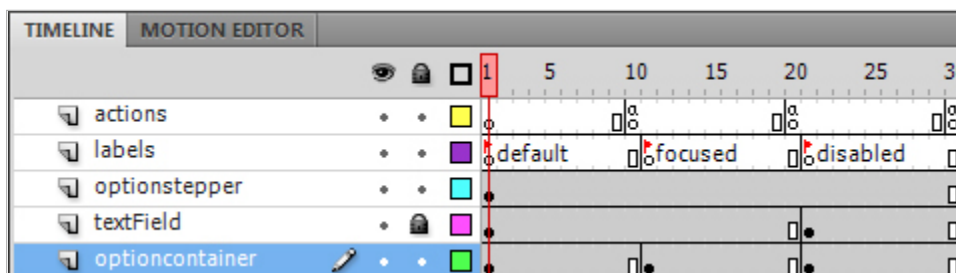
:

- **textField** TextField
- **nextBtn** CLIK Button dataProvider
- **prevBtn** CLIK Button dataProvider

2.4.2.3

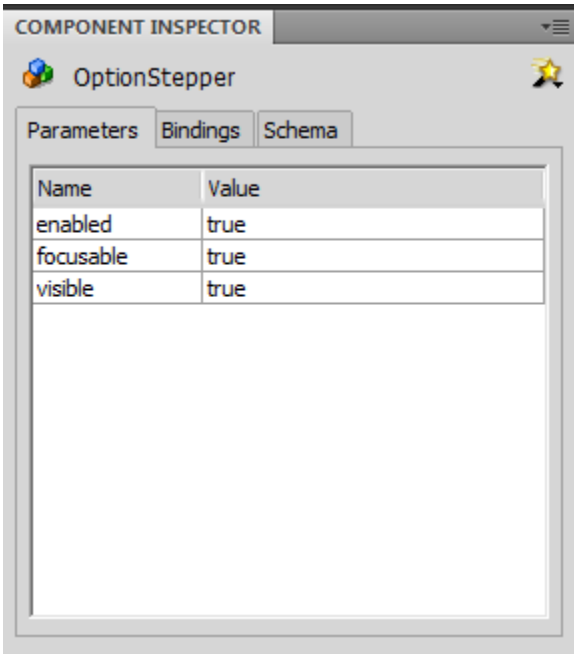
OptionStepper

- **default** enabled
- **focused** textField
- **disabled**



36: OptionStepper

2.4.2.4



OptionStepper		MovieClip	
Enabled	false,		
Focusable	/ tab	focusable()	false
Visible	false		

2.4.2.5

Event() Event ,

- *type*
- *target*
- *currentTarget* Event
- *eventPhase* EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- *bubbles*
- *cancelable*

OptionStepper

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false

ComponentEvent.STATE_CHANGE			
FocusHandlerEvent.FOCUS_IN			
FocusHandlerEvent.FOCUS_OUT			
IndexEvent.INDEX_CHANGE	OptionStepper		
	index:NumericStepper	int	-
	1(		1
	lastIndex:NumericStepper		int
	-1(		)
	1		
	data:	dataProvider	AS3 Object

OptionStepper

```
myOS.addEventListener(IndexEvent.INDEX_CHANGE, onValueChange);
function onValueChange(e:IndexEvent) {
    trace("myOS.selectedItem: " + e.target.selectedItem);
    // Do something
}
```

2.4.3 ListItemRenderer



2: **ListItemRenderer.**

ListItemRenderer (scaleform.clik.controls.ListItemRenderer) CLIK Button

ScrollingList TitleList DropDownMenu

2.4.3.1

ListItemRenderer Button
 ListItemRenderer
 ListItemRenderer

2.4.3.2

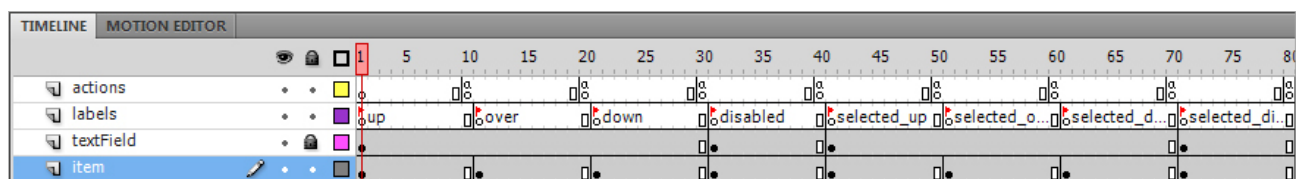
CLIK ListItemRenderer MovieClip :

- **textField** () TextField
- **focusIndicator**: () MovieClip MovieClip
 “ show ” “ hide ”

2.4.3.3

ListItemRenderer *selected*

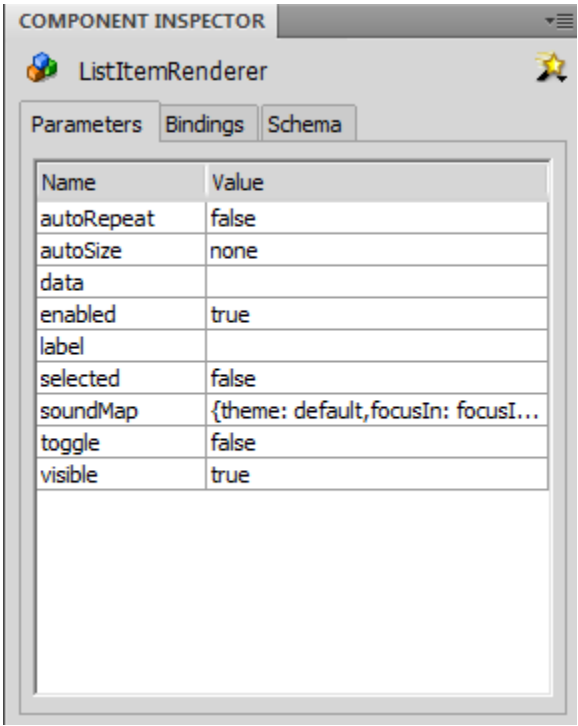
- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*



38: ListItemRenderer

ListItemRenderer Button
 ListItemRenderer CLIK

2.4.3.4



ListItemRender

autoRepeat	ListItemRender	"click" ()
autoSize	ListItemRender ListItemRender autoSize=TextFieldAutoSize.NONE	autoSize
Data	ListItemRender ListItemRender	List() (ScrollingList / TileList)
Enabled	false,	
Label	ListItemRender	
selected	ListItemRender	ListItemRender : ListItemRender toggle true , , , toggle false, ActionScript
Toggle	ListItemRender toggle	true, ListItemRender
Visible	false	

```
Event( ) Event ,
```

ListItemRenderer	ITEM_PRESS	ITEM_CLICK

EventListener	(ListEvent ITEM, C
---------------	--------------------

ListItemRenderer EventListener

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
Event.SELECT		
MouseEvent.ROLL_OVER	mousedx:) uint MouseEventEx buttonIdx: Scaleform, MouseEventEx	(Scaleform, ()
MouseEvent.ROLL_OUT	mousedx:) uint MouseEventEx buttonIdx: Scaleform, MouseEventEx	(Scaleform, ()

ButtonEvent.PRESS	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false – isRepeat: true – autoRepeating ; false –
MouseEvent.DOUBLE_CLICK	doubleClickEnabled true mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx
ButtonEvent.CLICK	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false – isRepeat: true – autoRepeating ; false –
ButtonEvent.DRAG_OVER	controllerIdx: () uint
ButtonEvent.DRAG_OUT	controllerIdx: () uint
ButtonEvent.RELEASE_OUTSIDE	controllerIdx: () uint

2.4.4 ScrollingList



39: ScrollingList.

ScrollingList (scaleform.clik.controls.ScrollingList)

ScrollIndicator

ScrollBar

dataProvider

```
scrollingList.dataProvider = new DataProvider(["item1", "item2", "item3", "item4"]);
```

crollingList

ListItemRenderer

FLA

2.4.4.1

ScrollBar

ScrollingList

ScrollingList

ScrollingList *wrapping*

wrapping

“ normal”

wrapping

“ wrap”

wrapping

“ stick”

(Page Up)

(Page down)

(Home)

(End)

ScrollBar

ScrollingList

ScrollaBar

D-Pad

CLIK

UI

2.4.4.2

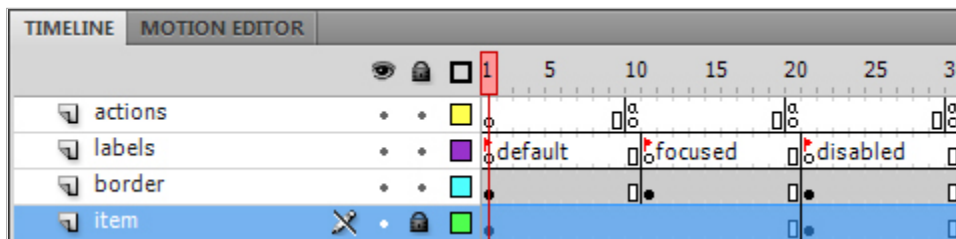
ScrollingList

ScrollingList

2.4.4.3

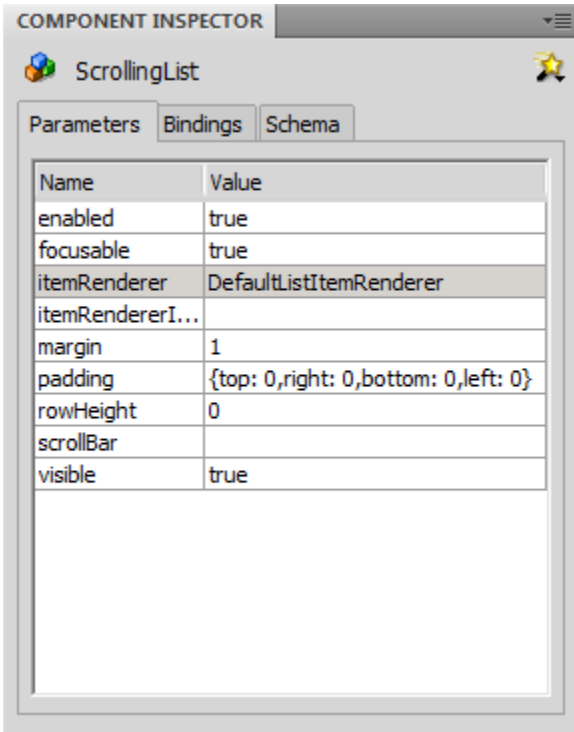
ScrollingList

- **default** enable
- **focused**
- **disabled**



40: ScrollingList

2.4.4.4



ScrollingList

MovieClip

Enabled	false,
Focusable	/ focusable() false tab
itemRenderer	ListItemRenderer itemRendererInstanceName (ListItemRenderers),
itemRendererInstanceName	ScrollingList ‘ r’ ‘ r1’ ‘ r2’ ‘ r3’ 1
Margin	rendererInstanceName
Padding	(Padding) itemRendererInstanceName (ListItemRenderer),

	ScrollBar()	
rowHeight	itemRendererInstanceName (ListItemRenderer),	
Visible	false,	ScrollBar
	ListItemRenderer	

2.4.4.5

Event() Event ,

- *type*
- *target*
- *currentTarget* Event
- *eventPhase* EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- *bubbles*
- *cancelable*

ScrollingList

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
ListEvent.INDEX_CHANGE		
	• itemRenderer: IListItemRenderer • itemData: dataProvider ActionScript Object • index: int -1() 1 • controllerIdx: /	

ListEvent.ITEM_PRESS

- ## ListEvent.ITEM_CLICK

- ## ListEvent.ITEM_DOUBLE_CLICK

- ## ListEvent.ITEM_ROLL_OVER

- 61

	- index: int -1() 1 - controllerIdx: / ()
ListEvent.ITEM_ROLL_OUT	- itemRenderer: IListItemRenderer - itemData: dataProvider ActionScript Object - index: int -1() 1 - controllerIdx: / ()

```

myList.addEventListener(ListEvent.ITEM_CLICK, onItemClick);
function onItemClick(e:ListEvent) {
    trace("ListItemRenderer Clicked: " + e.itemRenderer);
    // Do something
}

```

2.4.5 TileList



dataProvider

```
tileList.dataProvider = new DataProvider(["item1", "item2", "item3",  
"item4", "item5"]);
```

TileList

ListItemRenderer

FLA

2.4.5.1

ScrollBar

TileList

TileList

(Page Up) (Page down)
(Home) (End)

TileList

ScrollBar

ScrollBar

2.4.5.2

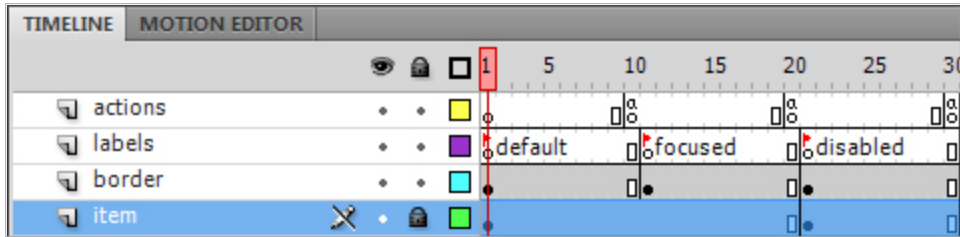
TileList

TileList

2.4.5.3

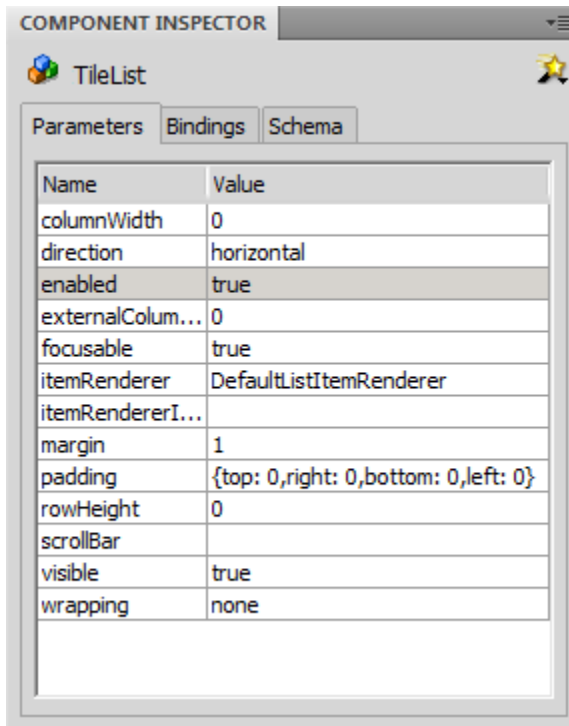
TileList

- **default** enable
- **focused**
- **disabled**



42: TileList

2.4.5.4



TileList

MovieClip

columnWidth	<i>rendererInstanceName</i>		
Direction			
Enabled	false,		
externalColumnCount	<i>rendererInstanceName</i>		
	TileList		
Focusable	/	focusable()	
	false	tab	

itemRenderer	ListItemRenderer itemRendererInstanceName (ListItemRenderer),
itemRendererInstanceName	ScrollingList Stage 'r', :'r1', 'r2', 'r3',... 1
Margin	itemRendererInstanceName (ListItemRenderer),
Padding	(Padding) itemRendererInstanceName (ListItemRenderer), ScrollBar()
rowHeight	itemRendererInstanceName (ListItemRenderer),
Visible	false, ScrollBar ListItemRenderer

2.4.5.5

Event() Event ,

- *type*
- *target*
- *currentTarget* Event
- *eventPhase* EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- *bubbles*
- *cancelable* TileList

ComponentEvent.SHOW	visible() true
ComponentEvent.HIDE	visible false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	

FocusHandlerEvent.FOCUS_OUT

ListEvent.INDEX_CHANGE

- itemRenderer:
IListItemRenderer
- itemData:
dataProvider ActionScript Object
- index: int -1(
- controllerIdx: /

ListEvent.ITEM_PRESS

- [illegible]

ListEvent.ITEM_CLICK

- itemRenderer:
IListItemRenderer
- itemData:
dataProvider ActionScript Object
- index:
int -1() 1
- controllerIdx:
() /

ListEvent.ITEM_DOUBLE_CLICK

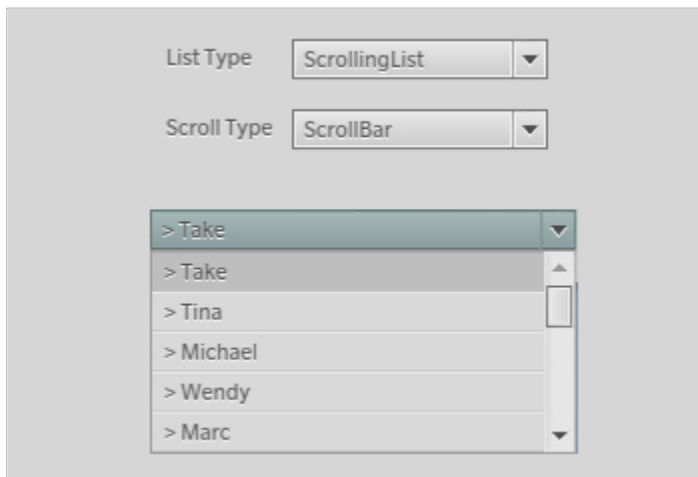
- itemRenderer:
IListItemRenderer
- itemData:

	dataProvider	AScript	Object
	- index: int -1() 1 - controllerIdx: / ()		
ListEvent.ITEM_ROLL_OVER	- itemRenderer: IListItemRenderer - itemData: dataProvider	AScript	Object
	- index: int -1() 1 - controllerIdx: / ()		
ListEvent.ITEM_ROLL_OUT	- itemRenderer: IListItemRenderer - itemData: dataProvider	AScript	Object
	- index: int -1() 1 - controllerIdx: / ()		

TileList

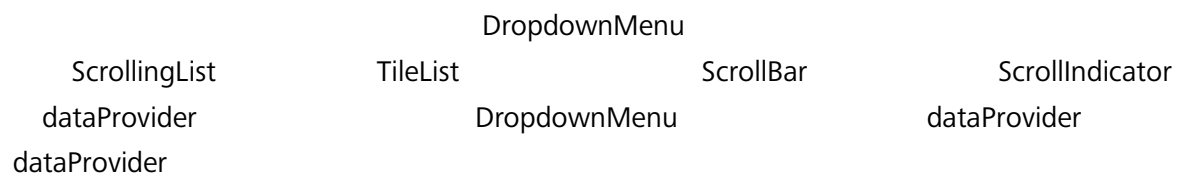
```
myList.addEventListener(FocusHandlerEvent.FOCUS_IN, onListFocused);
function onListFocused(e:FocusHandlerEvent) {
    trace("myList is now focused!");
    // Do something
}
```

2.4.6 DropdownMenu

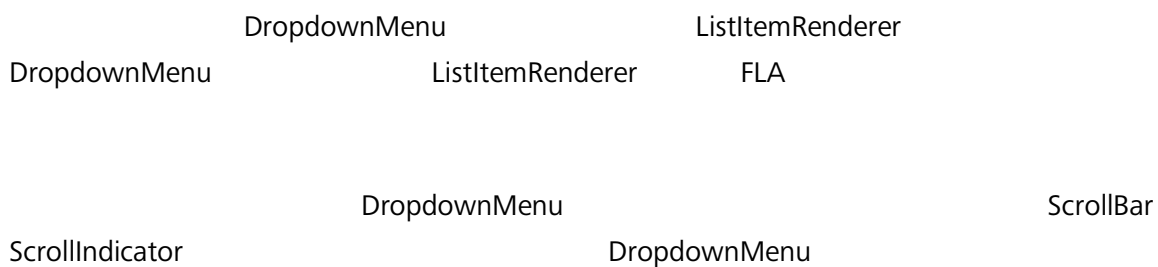


43: DropdownMenu.

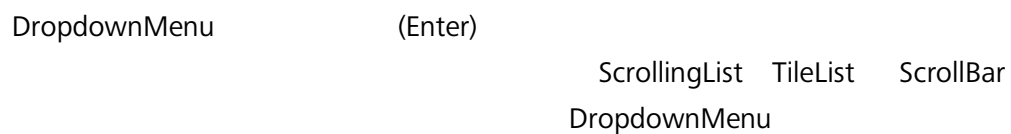
DropdownMenu (scaleform.clik.controls.DropdownMenu)



```
dropdownMenu.dataProvider = new DataProvider(["item1", "item2", "item3", "item4"]);
```



2.4.6.1



2.4.6.2

DropDownMenu

MovieClip

- **textField** () TextField
- **focusIndicator:** () MovieClip
"show" "hide"

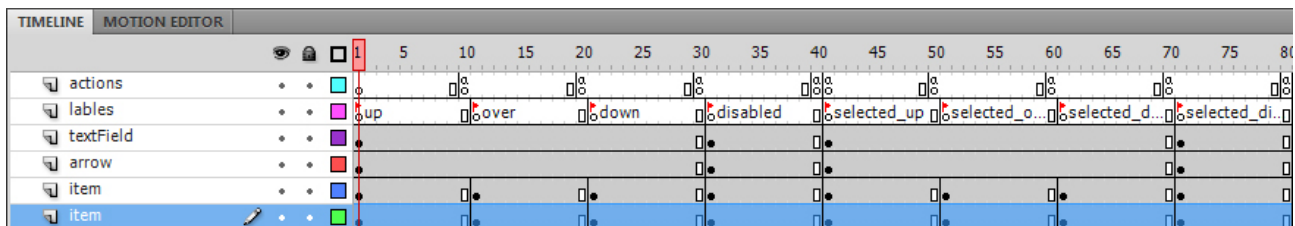
2.4.6.3

DropDownMenu

ToggleButton

CheckBox

- **up**
- **over**
- **down**
- **disabled**
- **selected_up**
- **selected_over**
- **selected_down**
- **selected_disabled**



44: DropDownMenu

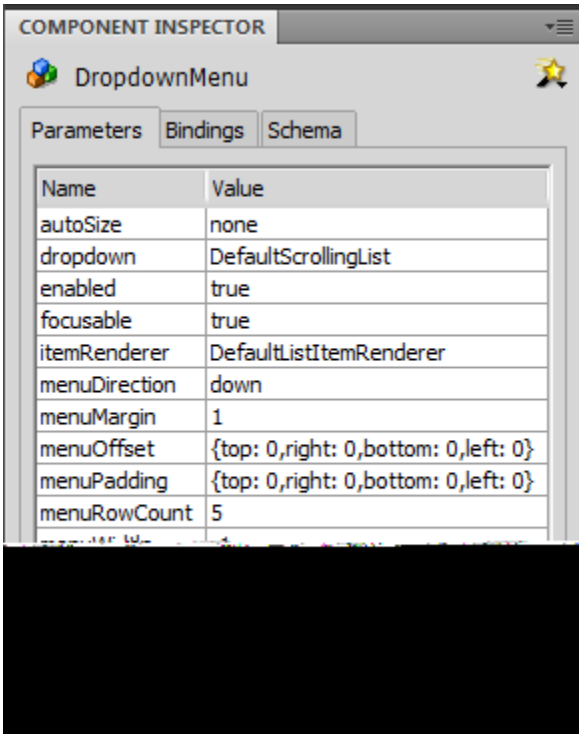
DropDownMenu

Button

DropDownMenu

[CLIK](#)

2.4.6.4



DropdownMenu

autoSize	DropdownMenu()	autoSize
	autoSize="none"	
Dropdown	ScrollingList TileList	
	DropdownMenu	
Enabled	false,	
Focusable	,	false
menuDirection	"up"() "down"()	
menuMargin		
menuOffset		
menuPadding		
menuRowCount		
menuWidth	If set, this number will be enforced as the width of the drop down's list.	

thumbOffset	,
scrollbar	,
Visible	false,

2.4.6.5

	Event()	Event	,
• <i>type</i> • <i>target</i> • <i>currentTarget</i> • <i>eventPhase</i> • <i>bubbles</i> • <i>cancelable</i>	Event	EventPhase.CAPTURING_PHASE	EventPhase.AT_TARGET
	EventPhase.BUBBLING_PHASE		
DropDownMenu	change	Button	

ComponentEvent.SHOW	visible()	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
Event.SELECT		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
ListEvent.INDEX_CHANGE	• itemRenderer: IListItemRenderer • itemData: dataProvider ActionScript Object • index: int -1() 1 • controllerIdx: /	

	(	)
MouseEvent.ROLL_OVER	mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx	
MouseEvent.ROLL_OUT	mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx	
ButtonEvent.PRESS	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false – isRepeat: true – autoRepeating ; false –	
MouseEvent.DOUBLE_CLICK	doubleClickEnabled true mouseIdx: () uint Scaleform, MouseEventEx buttonIdx: () Scaleform, MouseEventEx	
ButtonEvent.CLICK	controllerIdx: () uint isKeyboard: true - (Keyboard)/ (Gamepad) ; false – isRepeat: true – autoRepeating ; false –	
ButtonEvent.DRAG_OVER	controllerIdx: (	

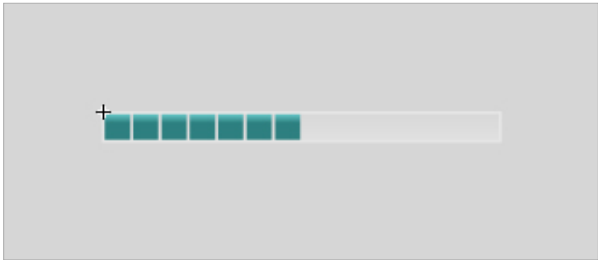
	) uint
ButtonEvent.DRAG_OUT	controllerIdx: (
	) uint
ButtonEvent.RELEASE_OUTSIDE	controllerIdx: (
	) uint

2.5



45: *Mercenaries 2* Faction

2.5.1 StatusIndicator



46: **StatusIndicator.**

StatusIndicator component (scaleform.clik.controls.StatusIndicator)

2.5.1.1

StatusIndicator

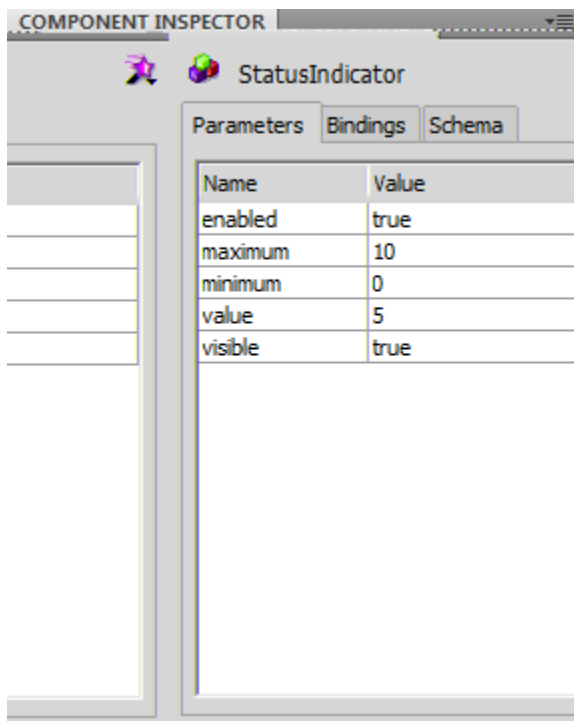
2.5.1.2

CLIK StatusIndicator MovieClip
 stop()
gotoAndStop()

2.5.1.3

StatusIndicator

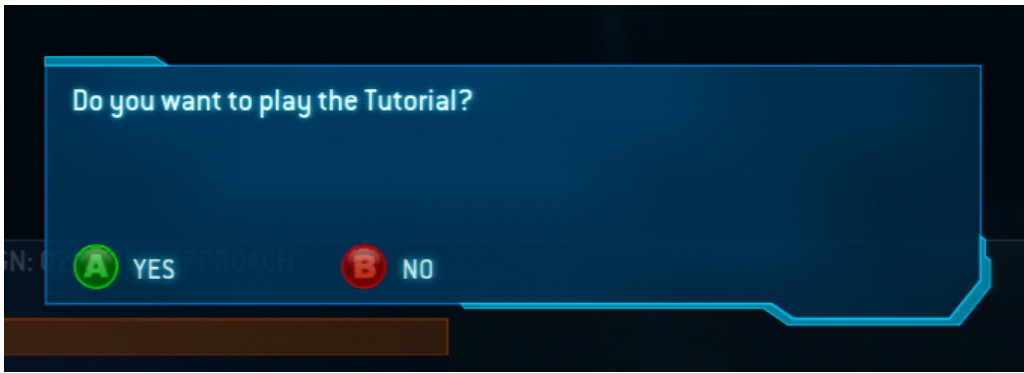
2.5.1.4



StatusIndicator

MovieClip

Visible	false	
Enabled	true	.



48: *Halo Wars* Window

2.6.1 Window

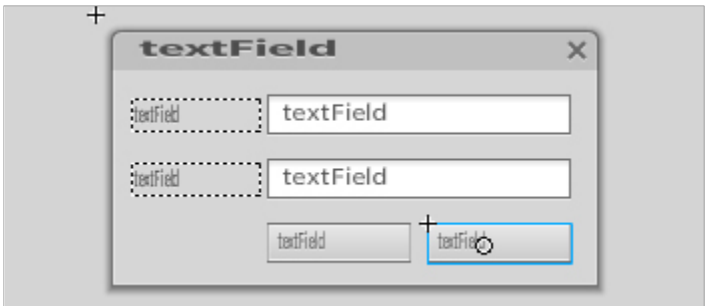


Figure 3: Sample unskinned Window.

CLIK Window (scaleform.clik.controls.Window) , Alert()

PopUpManager : (PopUpManager.showModal())

2.6.1.1

Window

2.6.1.2

CLIK Window MovieClip :

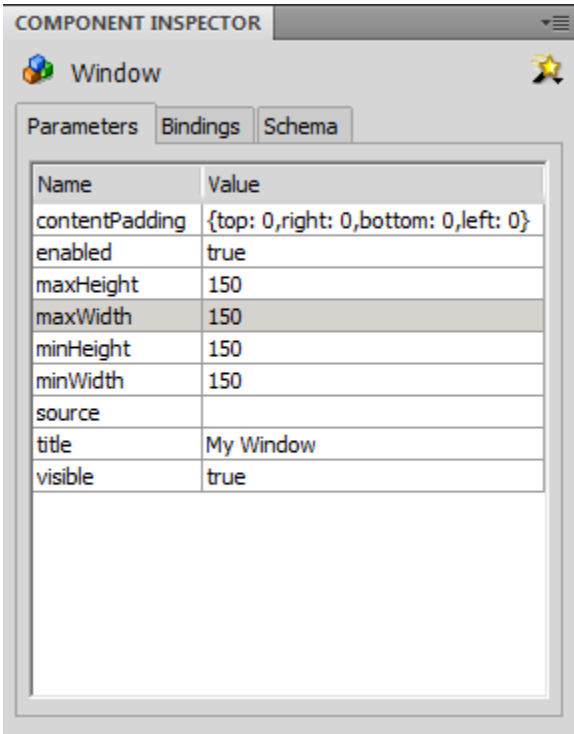
- **closeBtn** ()CLIK Button
- **titleBtn** ()CLIK Button

- **okBtn**)CLIK Button "Accept"() "OK"() ,
- **resizeBtn:**)CLIK Button , ,
- **background**)MovieClip ,
- **hit**)MovieClip hitArea

2.6.1.3

Window MovieClip

2.6.1.4



Dialog MovieClip

contentPadding		,
Enabled	true	.
maxHeight		
maxWidth		
minHeight		

minWidth	
Source	
Title	titleBtn ,
Visible	false

2.6.1.5

```

    Event( ) Event ,

```

- **type**
- **target**
- **currentTarget** Event
- **eventPhase** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET
EventPhase.BUBBLING_PHASE
- **bubbles**
- **cancelable**

Window

ComponentEvent.SHOW	visible	true
ComponentEvent.HIDE	visible	false

```

myWindow.addEventListener(ComponentEvent.HIDE, onWindowClosed);
function onWindowClosed(e:ComponentEvent) {
    // Do something.
}

```

3

Scaleform CLIK

3.1

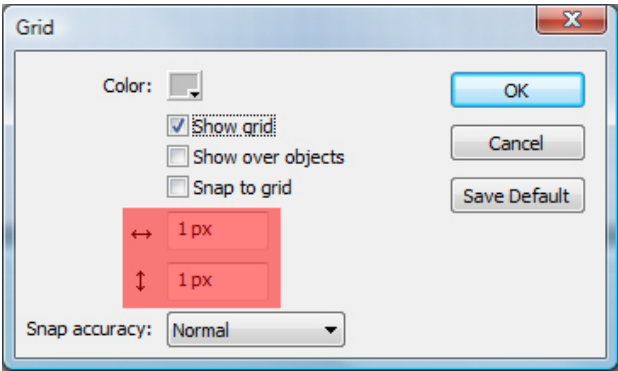
Scaleform CLIK

3.1.1

CLIK

Flash

- 1. Flash
- 2.
- 3. Flash
- 4.
- 5. 1px
- 6. OK

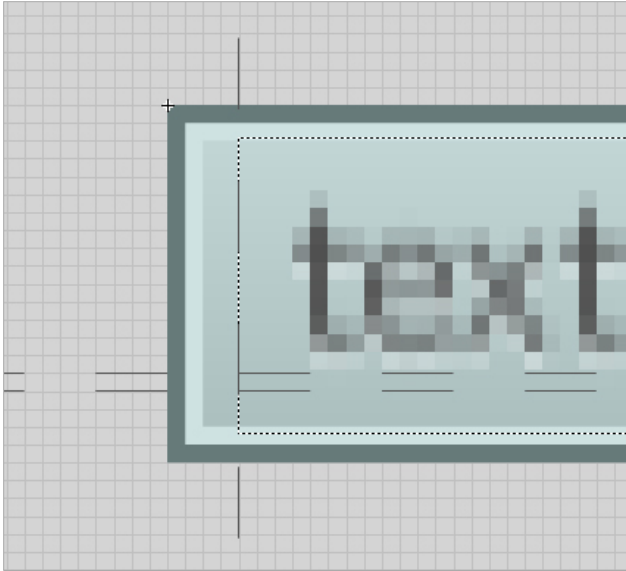


51: Grid

SWF

2

2



52:

3.1.2

Flash

Scaleform

Flash

Photoshop®

PNG

3.1.3

—

3.1.4

Flash

3.1.5

3.1.6 2

2 2

- 16x16
- 32x32
- 64x64
- 128x128
- 128x256
- 256x256
- 512x256
- 512x512

3.2

Flash

Scaleform CLIK

3.2.1

Flash

3.2.1.1

1

FLA

CLIK

FLA

1. *CLIK_Components.fla*

2. FLA

Copy

FLA *Paste*

3. FLA *Rename* ' Button'

4. FLA *Properties*

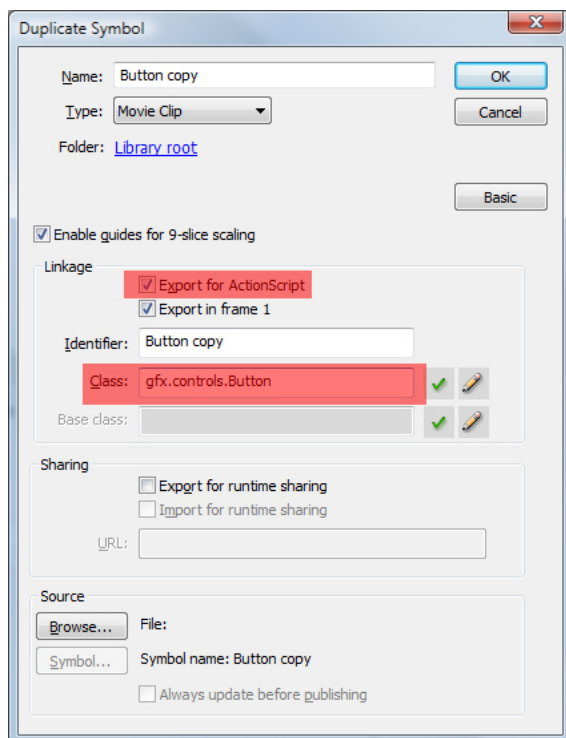
5. *Identifier* 3

6. *Paste*

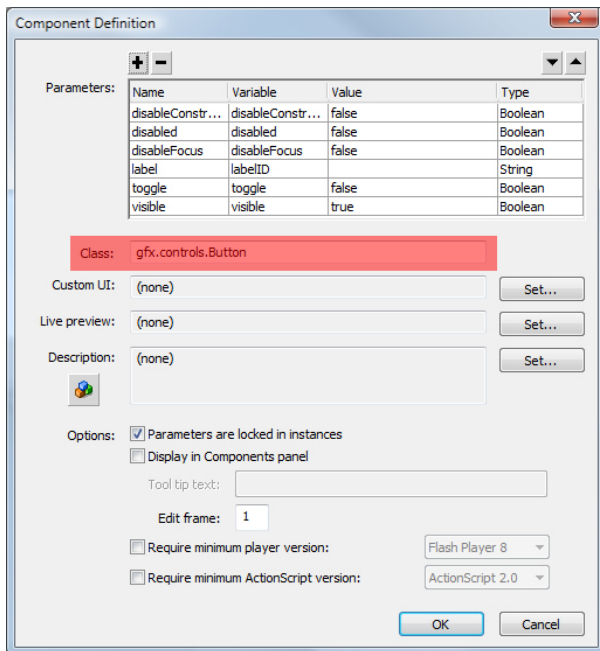
3.2.1.2

2

1. *library* *Properties*
2. *Properties* *scaleform.clik.controlsButton* *Class*
(CTRL+C)
3. *Cancel*
4. *Duplicate*
5. *Duplicate Symbol* *Export* *ActionScript*
6. *Class* (CTRL+V) *textField*
7. *OK*
8. *library* *Component Definition*
9. *Class* *textField* (CTRL+V) *textField*
10. *OK*



53:



54:

3.3

Scaleform CLIK

CLIK

CLIK

Flash

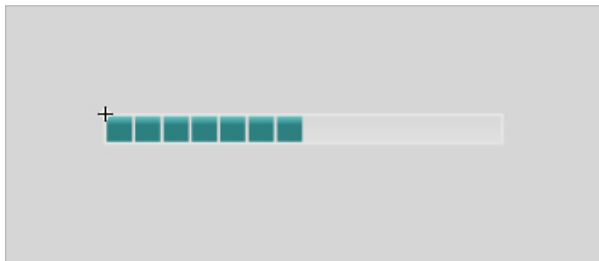
FLA

- Flash
- Flash
- Photoshop Illustrator®

Flash

CLIK

3.3.1 StatusIndicator



55: StatusIndicator.

StatusIndicator

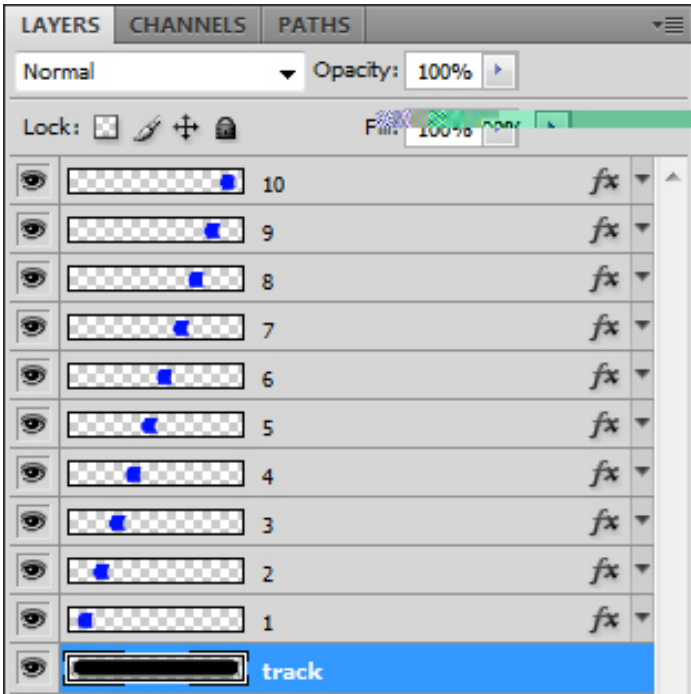
indicator track Track

Indicator

Photoshop

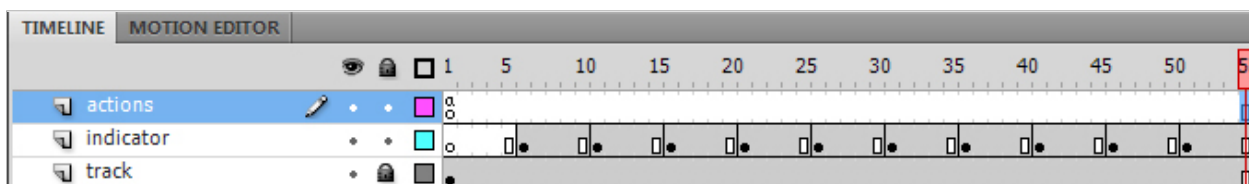


56: StatusIndicator PSD



57 Photoshop StatusIndicator PSD

1. Photoshop StatusIndicator
2. PSD
3. Flash *File Import -> Import to Stage*
4. StatusIndicator PSD
5. *Import Convert layers to: 'Layers'*
6. OK
7. Flash PSD 10
- PSD 10 1 10
- 1 10 *layer 1*
- 8.
- 9.
10. *layer 1* 6
11. *Insert layer 1* 11
12. *layer 2 (Ctrl+X)*
13. *layer 1* 11 (*Ctrl+Shift+V layer 1*)
14. 10 (*layer 1*)
- 5 *layer 2* 11
- layer 3* 16 *layer 4* 21
- 10 PSD 51
15. 2-10
- 16.
- Remove Frames*
- 55
17. *indicator layer*
18. *track track*
19. *layer 1 'indicator'*
20. *indicator track actions layer track indicator*

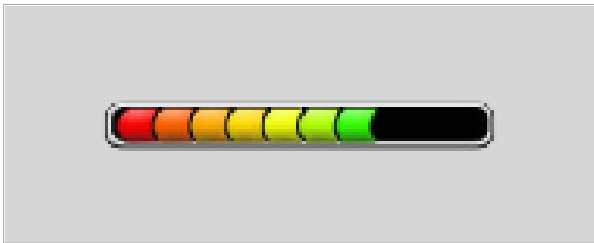


58:

21. StatusIndicator
22. 1 10

23.

24.



59: StatusIndicator

3.4

Scaleform CLIK

3.4.1

Scaleform

SWF

Scaleform

Flash UI

3.4.2

Flash

Scaleform

Scaleform

Scaleform

textFieldId

Slate Mobile

Slate

ASCII

Mobile

Slate Mobile

CJK

CJK

textFields

Scaleform

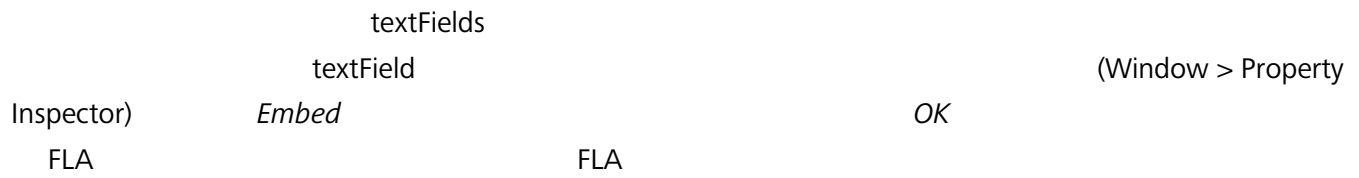
3.4.4

Scaleform

Scaleform

UI

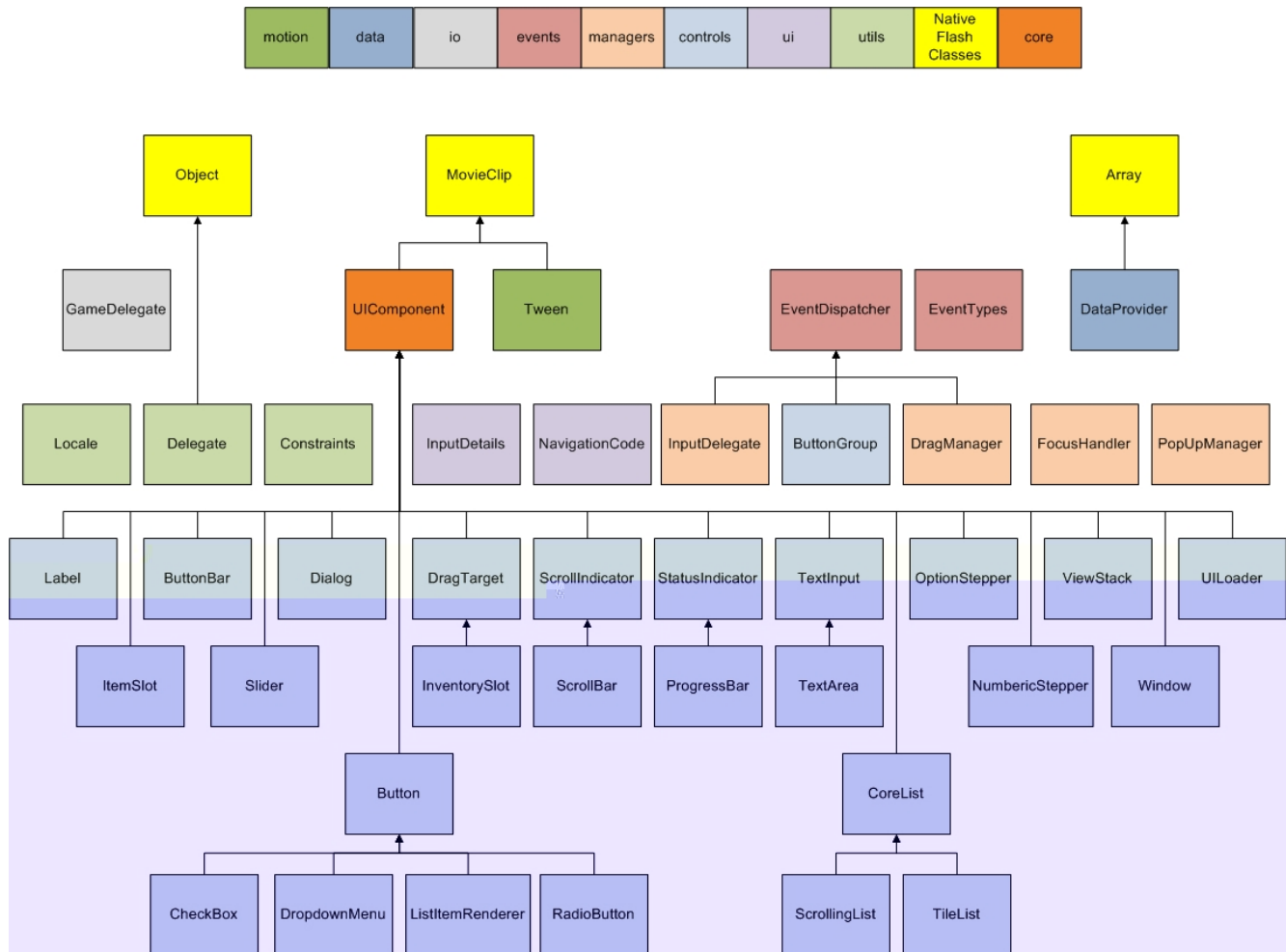
3.4.3 textField



3.4.4



Scaleform CLIK



60: Scaleform CLIK

4.1 UI *UIComponent*

Scaleform CLIK
(scaleform.clik.core.UIComponent).

UIComponent

UIComponent
CLIK
Scaleform

UIComponent

Flash 8

MovieClip

Flash

MovieClip

/

UIComponent

- **enabled;**
- **visible;**

- ***focused;***
- ***focusable;***
- ***width;***
- ***height;***
- ***scaleX;***
- ***scaleY;***
- ***displayFocus:*** true
- ***focusTarget:*** Settings this property to another MovieClip will cause that MovieClip to receive focus rather than this component if focus is ever set to this MovieClip. For more information see the [Focus Handling](#) section.

UIComponent

- ***configUI:***
- ***draw:***
- ***changeFocus:***
- ***scrollWheel:***

UIComponent EventDispatcher

4.1.1

- onLoad()
- MovieClip
- configUI()
- validateNow()
- draw()

4.2

Scaleform CLIK

4.2.1

ListItemRenderer RadioButton CheckBox CLIK Button
 Button CLIK Button (scaleform.clik.controls.Button)
 Button ScrollBar

- *up*
-
-
- *disabled*

Button

“selected_”

CLIK

Button

- *up*
-
-
- *disabled*
- *selected_up*
-
-
- *selected_disabled*

down

over

selected_over

selected_down

CLIK

CLIK

CLIK

getStatePrefixes()

```
protected function getStatePrefixes():Vector.<String> {
    return _selected ? statesSelected : statesDefault;
}
```

CLIK

“selected_”

getStatePrefixes()

stateMap

CLIK

```
protected var _stateMap:Object = {
    up: ["up"],
```

```

        over: ["over"],
        down: ["down"],
        release: ["release", "over"],
        out: ["out", "up"],
        disabled: ["disabled"],
        selecting: ["selecting", "over"],
        toggle: ["toggle", "up"],
        kb_selecting: ["kb_selecting", "up"],
        kb_release: ["kb_release", "out", "up"],
        kb_down: ["kb_down", "down"]
    }

```

getStatePrefixes()

61:

4.2.2

CLIK

ScrollingList OptionStepper Slider TextArea

- *default*
- *focused*
- *disabled*

4.2.3

- *default*
- *disabled*

4.2.4

	StatusIndicator	ProgressBar
0	50%	50%
10	gotoAndStop()	
5	updateValue()	
25		
50		

4.3

Scaleform CLIK event model “ ”

CLIK ActionScript 3

ActionScript 3 , :

http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/Event.html

4.3.1

4.3.1.1

```

    addEventListener()
    removeEventListener()
    ,
    CLIK
    Event(    )
    CLIK
    scaleform.clik.events
    ,addEventListener()
    ,:useCapture    priority
    useWeakReference
    ,

```

addEventListener()

```
buttonInstance.addEventListener(ButtonEvent.CLICK, onClick, false, 0, true);
```

addEventListener()

[http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/EventDispatcher.html#addEventListener\(\)](http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/EventDispatcher.html#addEventListener())

```
buttonInstance.addEventListener(ButtonEvent.CLICK, onClick, false, 0, true);
function onClick (e:Event):void {
    buttonInstance.removeEventListener(ButtonEvent.CLICK,
onClick, false);
}
```

, , buttonInstance ButtonEvent.CLICK
onClick
Event() CLIK

4.3.1.2

CLIK dispatchEvent()

```
dispatchEvent(new Event(Event.CHANGE));
```

4.4

ActionScript 3 , ,

```
import flash.utils.getDefinitionByName;
import scaleform.clik.controls.Button;
```

```
public var myButton:Button;
```

```
// Class
```


FocusHandler

UIComponent.changeFocus()

```
function changeFocus():void {
    track.displayFocus = _focused;
}
```

false

```
background.tabEnabled = false;
```

4.5.2

Selection

3. displayFocus true changeFocus

4.6

Scaleform CLIK

Flash 8 MovieClip

MovieClip

CLIK

4.6.1

```

class (scaleform.clik.managers.InputDelegate)
    InputDetails (scaleform.clik.ui.InputDetails)
    InputDelegate.readInput()
    InputDetails
    InputDelegate

```

InputDetails

- “ key ”
-
-

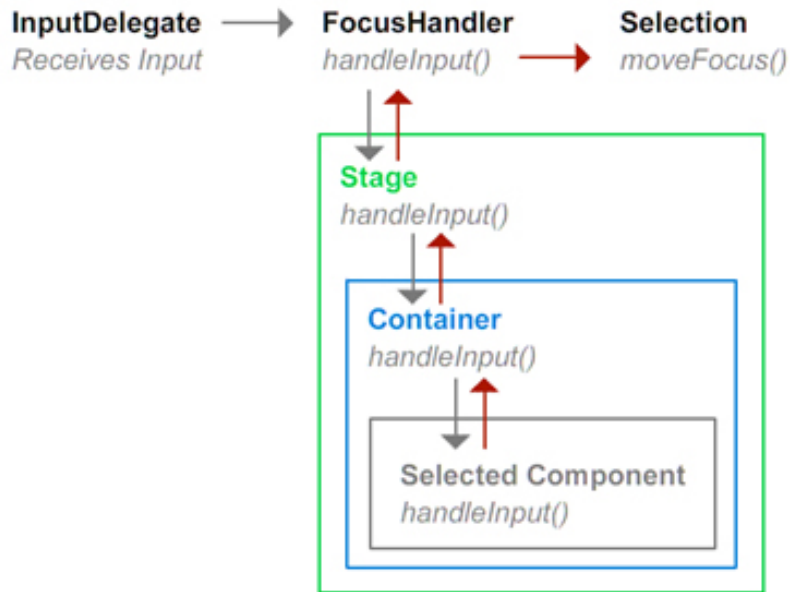
- “ InputValue.KEY_UP (“keyUp”) InputValue.KEY_DOWN (“keyDown”)
 - navEquivalent(navigation equivalent) “as NavigationCode.UP (“up”),
 - NavigationCode.DOWN (“down”), NavigationCode.LEFT (“left”), NavigationCode.RIGHT (“right”),
 - NavigationCode (scaleform.clik.ui.NavigationCode)

```

FocusHandler    InputDelegate    InputEvent    ,
InputEvent,    bubble

```

the handleInput()



62: handleInput()

```

    InputEvent      ,handled      true
event.isDefaultPrevented(),

```

```

override public function handleInput(event:InputEvent):void {
    // Check whether the event has already been handled by another component.
    if (event.isDefaultPrevented()) { return; }
    var details:InputDetails = event.details;

    switch (details.navEquivalent) {
        case NavigationCode.ENTER:
            if (details.value == InputValue.KEY_DOWN) {
                // Do something.
                event.handled = true;
            }
            break;
        default:
            break;
    }
}

```

```

InputEvent      ActionScript 3      (bubble),      .bubble
ScrollingList,      InputEvent
Enter ,ListItemRenderer      ButtonEvent.CLICK

```

```

override public function handleInput(event:InputEvent):void {
    if (event.isDefaultPrevented()) { return; }
    // Pass on to selected renderer first

```



```

    var renderer:IListItemRenderer = getRendererAt(_selectedIndex,
_scrollPosition);
    if (renderer != null) {
        // Since we are just passing on the event, it won't bubble, and should
properly stopPropagation.
        renderer.handleInput(event);
        if (event.handled) { return; }
    }
    . . .
}

```

InputDelegate.instance

FocusHandler

```

InputDelegate.instance.addEventListener(InputEvent.INPUT, handleInput);
function handleInput(e:InputEvent):void {
    var details:InputDetails = e.details;
    if (details.value == Key.TAB) { doSomething(); }
}

```

InputDelegate

InputDelegate

W A S D

4.6.2

Nintendo Wii™

CLIK

SWF

```

myButton.addEventListener(ButtonEvent.CLICK, onCursorClick);
function onCursorClick(event:ButtonEvent):void {
    trace(event.controllerIdx);
}

```

4.7

```

btnInstance.setSize(100,22);
btnInstance.width = 200;
btnInstance.label = "text";
draw();
btnInstance.toggle = true;
btnInstance.selected = true;
btnInstance.data = obj;
invalidate();

```

63: CLIK

4.7.1

```

UIComponent.draw()
invalidate()
UIComponent.invalidate()
draw()

,draw()
isInvalid()
( , )
draw()
invalidate()
isInvalid() draw()
( invalidateData(), invalidateSize())
invalidate()
(Event.RENDER)
draw()
(Event.ENTER_FRAME)
draw()
UIComponent.validateNow()
draw()

```

4.8

Scaleform CLIK

1.

2.

scale9grid

Flash scale9Grid , “ ” ,
Flash/Flex , container

CLIK , Scaleform scale9Grid
CLIK ,
container () , ,

Flash IDE setSize()
Flash IDE
MovieClips LivePreviews Scaleform Player
CLIK

4.8.1 Scale9Grid

Scaleform Player Scale9Grid MovieClip
scale9grid Flash MovieClips scale9grid Flash
Player Scaleform
MovieClip scale9grid
Scale9grid

4.8.2

Constraints (scaleform.clik.utils.Constraints)
ScrollBar
Constraints
ScrollBar initialize()
draw()
draw()

```
override protected function initialize():void {
    super.initialize();

    // The upArrow doesn't need a constraints, since it sticks to top left.
    if (downArrow) {
        constraints.addElement("downArrow", downArrow, Constraints.BOTTOM);
    }
}
```

```

    }
    constraints.addElement("track", track, Constraints.TOP |
                          Constraints.BOTTOM);
}

override protected function preInitialize():void {
    constraints = new Constraints(this, ConstrainMode.REFLOW);
}

protected function draw():void {
    constraints.update(_width, _height);
}

```

4.9

dataProvider dataProvider
 Scaleform CLIK IDataProvider (scaleform.clik.interfaces.IDataProvider)

API

dataProvider

dataProvider CoreList (ScrollingList, TileList) OptionStepper

DropDownMenu

4.9.1

DataProvider (scaleform.clik.data.DataProvider) initialize()

dataProvider ActionScript dataProvider

dataProvider

dataProvider IDataProvider

```

myComponent.dataProvider = new DataProvider([
    "data1",
    4.3,
    {label:"anObjectElement", value:6}]);

```

dataProvider

- length:

- *requestItemAt:* dataProvider
OptionStepper
- *requestItemRange:* dataProvider
ScrollingList
- *indexOf:*
- *invalidate:* dataProvider
“ dataChange ” dataProvider

dataProvider

ActionScript 2

dataProvider ExternalInterface.call

4.10

Scaleform CLIK Tween (scaleform.clik.motion.Tween), Tween
, Scaleform CLIK Tween , fl.transitions.easing.*

tween, Tween(), () Sprite tween
tween , Tween /

- *ease* Function()
- *easeParam* Object()
- *onComplete* Function Tween (closure)
- *onChange* Function Tween (/ (tick/frame))
- *data* Object Tween (Tween)
- *nextTween* Tween Tween Tween ,
Tween *pause=false*
- *frameBased* Boolean ,
- *delay* Number tween x
- *fastTransform* Boolean
- *paused* Boolean Tween

Tween Tween MovieClip Tween Tween
Tween MovieClip
Tweens

4.10.1

Tween MovieClip MovieClip tween

 tween , , onComplete
 Tween Tween ,
 Tween

```
import fl.transitions.easing.*;
import scaleform.clcik.motion.Tween;

// Perform a 1 second tween from the current horizontal position and
// alpha values to the ones specified
var myTween:Tween = new Tween(300,
                               myMovieClip,
                               { x:100, y:100, alpha:0 },
                               { ease:Strong.easeOut,
                                 onComplete:onCompleteCallback} );

function onCompleteCallback(t:Tween):void { // Do something. }
```

4.11 佈局框架

CLIK
 ,CLIK UI, CLIK

:Scaleform v4.0.14 CLIK AS3 , API
 CLIK AS3 , Scaleform

CLIK

- **scaleform.clcik.layout.Layout** Sprite , Sprite
 .layoutData
- **scaleform.clcik.layout.LayoutData** Sprite

CLIK

1. Sprite/MovieClip Stage DisplayObjectContainer
2. LayoutData Sprite/MovieClip .layoutData

```

3.          Sprite/MovieClip  LayoutData          Layout
4.          1      Sprites/MovieClips          DisplayObjectContainer

,      Layout(      )          ,      Layout          Sprite
MovieClip,      Sprite      MovieClip          (x      y          ),
.layoutData

```

```

Scaleform\Resources\AS3\CLIK\demos\          SDK
Layout_Main      CLIK          ,          Scaleform SDK          Layout_Main
      MovieClip          Layout_FullScreen_Demo
FxPlayer          ,          Stage          Layout_FullScreen_Demo
          Layout

```

4.11.1

```

(Layout)          ,          /
      Sprite(          )          ,
,
(myWindow.myMovieClip      myWindow.layoutInstance)          layoutInstance      myMovieClip

      Sprite,      Sprite      LayoutData          .layoutData
,          ,          .identifier          ,          Sprite          LayoutData
.layoutIdentifier          Stage          ,          ,
.layoutData

,Layout          ,          ,          Stage
.layoutData,Layout          Sprite/MovieClip          ,
Sprite/MovieClip          Stage
,          ,

```

public function reset():void

```

      Sprite          LayoutData      Sprite          /
Sprite,          Layout

```

public function resortManagedSprites():void

```

layoutData.layoutIndex          Sprite          ,
layoutIndex          ,

```

```

Stage Layout Flash (Symbol) ,
Layout Layout Symbol( 0 ),
Layout Layout .rect ,
,Layout , Sprite/MovieClip
LayoutData

```

4.11.1.1

Rect	" "	x y
	Layout != 0(Symbol, Stage), Layout	
	MovieClip x y Layout == 0(addChild()	
	Symbol), Layout x y .rect	
	Rectangle()	
tiedToStageSize	true – Layout Stage , false	
tiedToParent	true – Layout , false	
Hidden	true - Layout, false Layout	
	, visible = false	

4.11.2 LayoutData

Sprite scaleform.clik.layout.Layout

4.11.2. 1

alignH	Sprite LayoutMode.ALIGN_NONE
	LayoutMode.ALIGN_LEFT LayoutMode.ALIGN_RIGHT
alignV	Sprite LayoutMode.ALIGN_NONE
	LayoutMode.TOP LayoutMode.BOTTOM.property
	TextFieldAutoSize.NONE
offsetH	Layout relativeToH Sprite
	(-1),Layout Stage Layout/Sprite
	()
offsetV	Layout relativeToH Sprite
	(-1),Layout Stage Layout/Sprite
	()

offsetHashH	(Hash Table), Layout Stage (Layout.tiedToStageSize == true), , Layout Stage , 4:3,"LayoutData.offsetHashH[LayoutData.ASPECT_RATIO_4_3] = 70;" Sprite 70px
offsetHashV	(Hash Table), Layout Stage (Layout.tiedToStageSize == true), , Layout Stage , 4:3,"LayoutData.offsetHashV[LayoutData.ASPECT_RATIO_4_3] = 70;" Sprite 70px
relativeToH	Sprite Sprite null,Sprite Layout
relativeToV	此 Sprite 應與其垂直相關的 Sprite 的實例名稱。假如保留為 null,Sprite 就會相對於 Layout 而對齊。
layoutIndex	Layout Sprite Sprite – relativeToH relativeToV Sprite Sprite Layout layoutIndex (-1),
layoutIdentifier	LayoutData Layout (DisplayObjectContainer Layout) (null), Layout () Layout

4.12

Scaleform CLIK PopUpManager (scaleform.clik.managers.PopUpManager)

4.12.1

PopUpManager ID
initProperties , PopUp
x y relativeTo ,Sprite ,
Sprite

```
import scaleform.clik.managers.PopUpManager;
PopUpManager.createPopUp("PopupLinkageID", {alpha:.5}, 100, 100
myMovieClip);
```

```

Scaleform      InteractiveObjectEx.setTopmostLevel()
                                     root
                                     SWF

                                     topmostLevel

InteractiveObjectEx.setTopmostLevel()
                                     Z
                                     topmostLevel

true

```

4.13

```

DragManager      (scaleform.clik.managers.DragManager)

```

```

                                     ,
                                     DragEvent.DRAG_STATE
DragManager.handleStartDragEvent()
                                     ,
DragManager      ,
MouseEvent.MOUSE_UP
DragManager.handleEndDragEvent()

```

4.13.1



64: DragDemo

DragManager
DragManager

IDragSlot

IDragSlot

DragManager

Scaleform
UIComponent
DragSlot
DragSlot

IDragSlot
CLIK

DragSlot
/ /Bitmap
DragSlot

scaleform.clik.controls.DragSlot)
DragManager
DragSlot
IDragSlot

DragSlot

DragSlot

DragSlot
DragManager dragBegin dragEnd

DragSlot

DragSlot

DragSlot UI

5

Scaleform CLIK

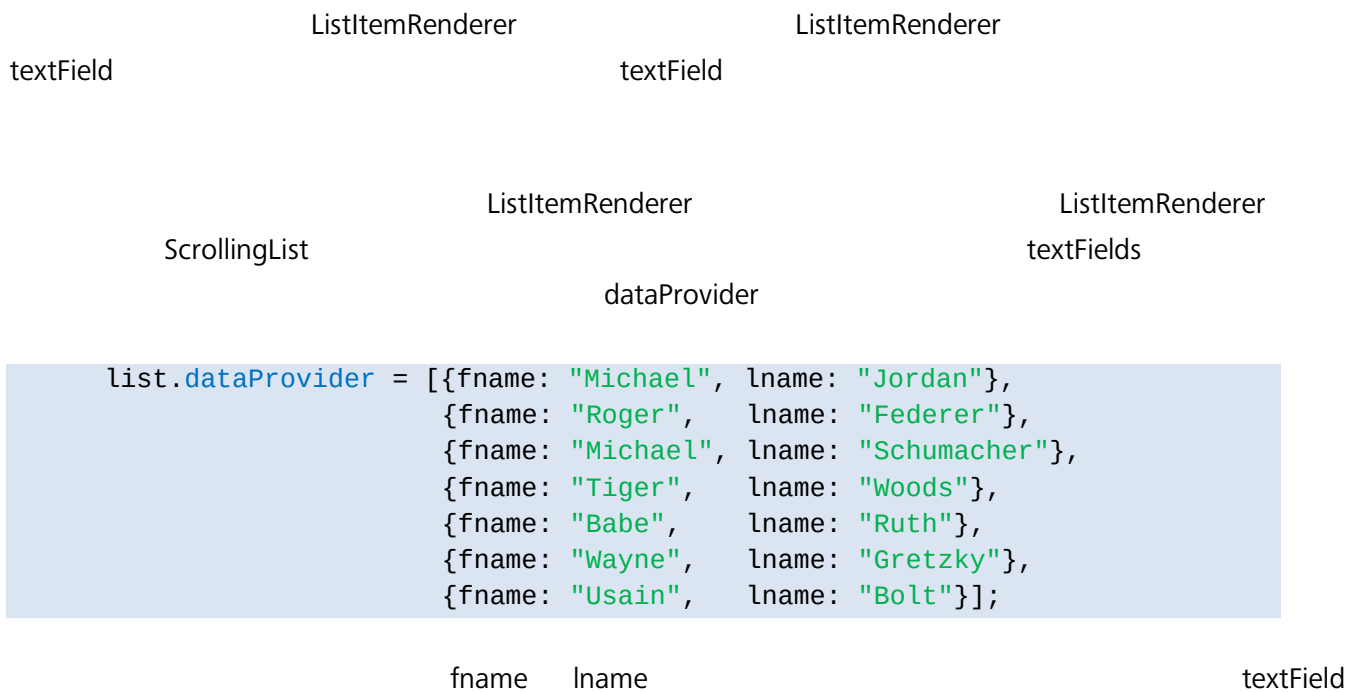
5.1

Scaleform CLIK

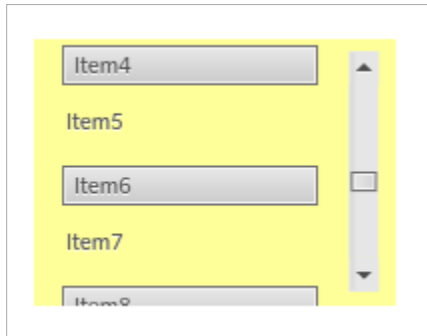
5.1.1 包含兩個 `textField` 的 `Listltem` `Renderer`



65: ScrollingList



5.1.2 圖元滾動視圖



66:

CLIK

CLIK

ScrollBar

- **actions:** ActionScript
- **scrollbar:** CLIK ScrollBar ' sb '
- **mask** MovieClip
- **content**
- **background**

' content' y 0

ActionScript

code

```
// 133 is the view size (height of the mask)
sb.setScrollProperties(1, 0, content.height - 133);
sb.position = 0;

sb.addEventListener(Event.SCROLL, onScroll, false, 0, true);
function onScroll(e:Event) {
    content.y = -sb.position;
}
```

setScrollProperties

6 常見問題解答

1.

1152: A conflict exists with inherited definition scaleform.clik...
in namespace public.
scaleform.clik...
public

“Automatically Declare Stage Instances”(ActionScript .FLA .FLA .FLA
),

CLIK CLIK , Stage ,
ActionScript

File -> Publish Settings -> "Flash" tab -> Actionscript 3.0 Settings
“Automatically declare stage instances”, .FLA "Automatically Declare Stage Instances"
, Flash Studio , .FLA

2. .FLA CLIK Stage
(inspectable) ,

1046: Type was not found or was not a compile-time constant:
...
,

-
- ActionScript
- ActionScript , – ActionScript
 ,myButton:Button (scaleform.clik.controls.), myButton:MyButtonSymbol,
 ,MyButtonSymbol , myButton

3. , .SWF ,

Symbol ‘MySymbol’, Layer ‘actions’, Frame 10, Line 1 -- 1024: Overriding a function that is not marked for override.


```
Symbol 'MySymbol', Layer 'actions', Frame 30, Line 1 -- 1024: Overriding a function
that is not marked for override.
```

```
"scaleform.clik.controls.Button"
```

7 潛在缺陷

getDefinitionByName()

8 CLIK AS3 vs. CLIK AS2

This section outlines major differences, mostly programming related, between CLIK AS2 and CLIK AS3.

1. CLIK AS3 is designed to work in Flash Player.
2. Rather than using a custom EventDispatcher, CLIK AS3 uses ActionScript 3's native event system.
3. The invalidation system has been redesigned to help avoid unnecessary updates in draw(). Each component now stores an "invalid" table that tracks which aspects of the component are currently invalid. Default invalidation types are defined in `scaleform.clik.constants.InvalidationType`. Aspects can be marked as invalid using `invalidate()`, `invalidate[InvalidationType]()`, or manually setting the property to true within the table.. Logic within draw() should be wrapped within an appropriate `isInvalid()` check to ensure that unnecessary updates to components are avoided.
4. `invalidate()` no longer uses a 1ms delay; instead, it uses the next stage invalidation (`Event.RENDER`) or the next `Event.ENTER_FRAME` for the component and then calls `validateNow()`.
5. Tween's syntax is no longer `MovieClip.tweenTo`; instead, `var t:Tween = new Tween();`
6. `handleInput()` is now tied to the native event system. This means that `handleInput()` now accepts one parameter of type `InputEvent` and the event will bubble up from the component that dispatched it rather than being passed down from `FocusHandler`.
7. `handleInput()` no longer returns a Boolean; instead, it should set `event.handled = true`; if the event was handled.
8. `UIComponent` has a new property, `focusable`, which can be used with the CLIK `FocusManager` to prevent focus from reaching a component. This is more or less a replacement for AS2's `MovieClip.focusEnabled` property.
9. `enableInitCallback` is no longer an inspectable of `UIComponent`. Instead, it should be set within a class's constructor or `preInitialize()` method.
10. `DataProvider` now requires that you provide the target array to `DataProvider`'s constructor (`myComponent.dataProvider = new DataProvider(myArray);`), rather than simply setting `myComponent.dataProvider = myArray`;
11. `ButtonBar` will now only create Buttons that fit within its own size. If your `ButtonBar`'s size is 200px wide and each Button is 150px wide, only one Button will be displayed. This allows users to resize their `ButtonBar` and have the Buttons within be added and removed dynamically.
12. `UILoader` has been removed. This functionality is now handled by the Flash `MovieClip/Sprite` and `Loader` classes.
13. `UIComponent.SoundMap`, introduced in CLIK 3.3, has been removed for CLIK AS3.